Almost Dead

Project Team

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Chapter 1

Introduction

"Almost Dead" is a groundbreaking VR zombie survival game that shifts away from the typical combat-focused approach, offering a unique experience centered around strategic decision-making and escape. Set in a Pakistani university during a sudden zombie virus outbreak, the game challenges players to solve puzzles, manage resources, and navigate through crises rather than relying on combat. This innovative approach not only provides a fresh and intellectually stimulating gameplay experience but also highlights a culturally relevant setting, contributing to the growth of the gaming industry in Pakistan.

Our game, Almost Dead, is not just your ordinary zombie game; it has a much more edge-cutting approach, that is, it does not focus on combat-focused strategy but gives you a distinct experience revolving around escape and decision-making and taking strategy. Our game dares the players to utilize and manage resources and find ways through crises instead of solely relying on combat strategies. In order to propel the gaming industry in Pakistan, this inventive approach not only provides the innovative and thought-provoking gaming experience, but it also promotes a setting that is culturally suitable.

Chapter 2

Project Vision

2.1 Problem Statement

Most of the existing games greatly focus on shooting and survival as their main survival method, despite the fact that VR games are gaining massive popularity, overlooking the possibility for strategic problem-solving, non-violent, and escape models. These games mainly focus on combat and shooting and lack the zombie-themed VR games that focus on tactical human survival and escape through smart decision-making and problem-solving, especially in situations. For instance, one being stranded at a university during an unexpected virus outbreak. The existing games are behind paywalls, and they have limited combat strategies, mostly shooting and killing. This makes these games lacking in fine mechanics. Also, there is no such game that exists in an environment set in Pakistan on any Quest platform and also adds the element of a zombie outbreak scenario with prominence on critical thinking outbreaks for real-world crisis outbreaks.

2.2 Business Opportunity

With this game, we have a unique opportunity to:

- Profit from the rising demand for virtual reality experiences, which is expected to reach 92.31 billion by 2027.
- Adopting non-violent survival techniques will fill the gap in the VR game industry.
- In VR games with a zombie theme, put more emphasis on problem-solving and critical thinking than on combat.
- Draw in a broad spectrum of players, including those who favor cerebral puzzles over conventional combat-oriented games.

- Bring in a fresh setting (the Pakistani university) and target the regional market.
- Provide a strategic and immersive VR experience for free, without any paywalls.
- Provide a fun and engaging platform for players to learn about real-world crisis preparedness.

2.3 Objectives

- Create and execute a captivating three-dimensional environment that is modeled after a university in Pakistan, enabling players to explore various areas while avoiding zombie attacks.
- Provide game mechanics that prioritize resource management, problem-solving, and strategic thinking over combat. To overcome obstacles, players should rely on stealth, escape techniques, and puzzle-solving.
- Enable players to interact with the environment, non-playable characters (NPCs), and different in-game objects by giving them simple controls and an easy-to-use interface.
- Provide safe login and registration procedures for users by integrating Firebase Authentication, protecting their game progress and data.
- Create a competitive scoreboard that keeps track of players' survival times, goals completed, and tactical choices. This will encourage players to compete and feel a sense of accomplishment.
- In order to improve user experience and reduce motion sickness, optimize game performance to maintain a smooth frame rate (at least 60 FPS) during gameplay, particularly in virtual reality environments.
- To guarantee a smooth gaming experience, implement a system that enables players to save their game state, resume play, and track their progress across multiple levels.
- Incorporate captivating cut scenes into the gameplay to give players emotional depth and narrative context, drawing them into the narrative and strengthening their sense of immersion in the game world.
- To enable iterative improvements prior to the final release, conduct user testing sessions to collect feedback on gameplay mechanics, difficulty levels, and the overall player experience.

2.4 Project Scope

The scope of the project includes several key elements designed to create an immersive and engaging experience. First, the development will involve creating user authentication systems and displaying a leaderboard to enhance user interaction and competitiveness. The university environment will be built using 3D objects to provide a realistic setting. Non-violent survival and escape techniques will be utilized, offering a unique approach to gameplay that emphasizes strategy over combat. Furthermore, level progression will range from easy to hard, offering challenges that scale in difficulty. The game will promote critical thinking and crisis preparedness, pushing players to make strategic decisions. Lastly, dynamic sound effects, lighting, and AI integration for non-playable characters (NPCs) will be incorporated to heighten the realism and overall gaming experience.

2.5 Modules

2.5.1 Module 1: User Log Management

This module involves creating a back-end database with Firebase to store player logs. Analyzing these logs will help in making future updates and improvements to the game.

1. Back-end database creation with Firebase
2. Analysis of user logs for future updates

2.5.2 Module 2: Leader board

The leader board module will showcase player achievements, including rankings and scores. This system will highlight top players to motivate and encourage participation.

1. Display of player rankings and scores
2. Motivation through top player recognition

2.5.3 Module 3: UI/UX

This module is dedicated to designing the user interface and experience. Utilizing Figma, it aims to create visually appealing screens and interactive elements that align with the game's theme and enhance user interaction.

1. UI design using Figma
2. Creation of interactive and visually engaging prompts

2.5.4 Module 4: Cinematic

This module focuses on creating cinematic sequences that will serve as the introduction to the game, enhancing the story-line and establishing an emotional connection with the players. Cinematic will consist of scenes that represent the narrative of the game.

1. Introduction videos with cinematic shots
2. Story-line integration for emotional engagement

2.5.5 Module 5: Environment

The environment module involves designing game levels in alignment with the game's backstory. It will feature levels inspired by a zombie theme set in a university building and include the creation of relevant artifacts.

1. Level design based on a university building setting
2. Creation of zombie-themed artifacts

2.5.6 Module 6: VR Integration

To enhance player immersion, this module focuses on integrating VR features such as hand tracking and haptic feedback. It aims to optimize controls for a seamless VR experience.

1. Integration of hand tracking
2. Haptic feedback for enhanced immersion

2.5.7 Module 7: Mini-Map

This module involves implementing an interactive map within the game that displays the player's current location and nearby landmarks to assist in navigation and gameplay.

1. Interactive map showing player location
2. Display of nearby landmarks

2.5.8 Module 8: Game play Mechanics

This module defines the gameplay mechanics, including the representation of game levels as university building floors, exploration challenges, and progression through increasingly difficult levels.

1. Level design with progression through building floors
2. Survival challenges and item collection
3. Increasing difficulty with each level

2.5.9 Module 9: AI NPC's

The AI NPC's module focuses on creating artificial intelligence-driven non-playable characters that interact with players and enhance the overall gaming experience.

1. Development of AI-driven NPC's
2. Interaction mechanics to improve gameplay

2.6 Constraints

2.6.1 Platform Dependency

- The game has to be compatible with specific VR platforms (e.g., Oculus) and may require optimization to run smoothly on such devices.

2.6.2 Hardware Constraints

- The game's requirements for VR hardware—such as motion controllers, VR headsets, and maybe hand-tracking sensors—restrict players' access to it.
- In VR surroundings, the game has to keep a low latency and steady frame rate to avoid motion sickness.

2.6.3 Resources and Budget

- The project has to stay within the budget allotted, which includes paying for things like VR Oculus headsets.

- The time and resources available to the development team are constrained, which could have an impact on the size and scope of some features.

2.6.4 Development Deadlines

- The game must be completed within the academic or project timelines, focusing on core functionality to meet deadlines, potentially leaving out advanced features.

2.6.5 Technical Difficulty

- Technical issues that must be fixed in the allotted time may arise while implementing 3D Models and dynamic gameplay.

- To function well in a VR environment, the game's interactions and mechanisms, such as puzzle-solving and non-violent survival tactics, need to be carefully planned out

2.6.6 Security Requirements

- Proper configuration of Firebase Authentication is necessary to securely manage user data and protect player accounts from unauthorized access.

2.6.7 VR Optimization

- The game should ensure smooth transitions, fast loading times, and high-quality real-time 3D rendering without overloading system resources.

- Managing audio and visual assets, especially for dynamic lighting and sound, is essential to avoid performance drops.

2.6.8 UI Restrictions

- The VR environment imposes constraints on UI design, making it immersive and straightforward without overwhelming players.

- Elements like mini-maps and menus should be intractable in 3D to avoid disrupting gameplay.

2.6.9 Accessibility

- The game needs to be accessible to all skill levels, offering easy-to-understand controls and tutorials.
- Addressing concerns about motion sickness and physical disabilities will improve accessibility.

2.7 Stakeholder Description

In our project of game development in VR Technology, there are two stakeholders.

2.7.1 Developers

The main internal stakeholders are the developers of the project, who have shown their expertise and motivation throughout the process. They play a significant role as they have a deep understanding of the technicalities in the development of the project and ensure the quality of the project.

2.7.2 Users

Users are the players that will play our game with the expectation of good quality and simplicity as they will be directly affected by these factors.

2.7.3 Stakeholder Summary

Any modifications made during the deployment phase and throughout development will have an immediate impact on stakeholders.

2.7.3.1 Stakeholder 1

Name:Developers

Represent: Members of the FYP Team

Role and Expectations: Ensuring the product is developed with accuracy and efficacy is the main objective of the developers. They should make sure that the product can adapt to any modifications made in accordance with the specifications.

2.7.3.2 Stakeholder 2

Name: Users

Represents: Participants

Role and Expectations:Users are increasingly concerned with the quality of the product as well as the quality of service, as they should satisfy their expected standards. They anticipate that the new product will have an easy-to-use interface and a straightforward design.

2.7.4 Key High Level Goals and Problem Of Stakeholders

2.7.4.1 Key High Level Goals

Developers: Make a game that satisfies user demands and technological requirements. Make sure the game is easy to update and maintain.

Users: can anticipate a unique, strategic experience that is different from traditional fighting games.

2.7.4.2 Problem Of Stakeholders

Developers: The challenge for developers is to create a stable and distinctive game within strict technical and time constraints.

Users: seek out aesthetically pleasing, user-friendly games that emphasize strategic problem solving above conflict.

Chapter 3

SRS

3.1 List of Features

1. **User Authentication System**:

   Players will be able to register and log in securely. Players can create their accounts and log in with our User Authentication System to pick up right where they left off when playing the game.

2. **Leaderboard System**:

   Uniquely challenging each player for obtaining the high score via a series of survival time, strategic decision-making, and level completion.

3. **3D University Environment**:

   A Pakistani university, fully modeled in 3D for the first time.

4. **Non-Violent Survival**:

   Instead of combat, players have to rely on their escape strategies, solving complex puzzles, and critical decision-making in order to avoid or face the threat.

5. **Level Progression**:

   As players progress, levels will become more intricate with more complex puzzles and heightened survival scenarios.

6. **AI NPCs**:

   Non-playable characters (NPCs) react dynamically to player actions, introducing new challenges and variations in gameplay.

7. **Cinematic Cutscenes**:

Immersive, high-quality cinematic sequences that elevate the story and player connection.

8. **Dynamic Sound and Lighting**:

   Realistic lighting, 3D positional sound, and dynamic sound respond to player actions.

9. **Mini-Map**:

   A mini-map helps players understand the environment and track objectives.

3.2 Functional Requirements

- **User Authentication System:**
  - The system shall allow players to create an account and log in using their registered credentials.
  - Secure data storage shall be handled through Firebase authentication. Players shall also be able to resume from their last saved progress.
- **Leaderboard System:**
  - Players will be ranked by the system according to their level completion, survival time, and decisions.
- **3D University Environment:**
  - There will be interactive survival and escape challenges set in a fully textured 3D Pakistani university environment.
  - As the game advances, player shall be able to interact with different areas of university
- **Non-Violent Survival:**
  - Instead of having players fight, the game will use strategic mechanics to let players escape and survive.
  - Threats will be evaded by using obstacles and puzzles, with the increase in difficulty as the game advances.
- **Level Progression:**

  - The game shall increase in difficulty with more complex challenges as players complete the level as the game advances.
  - There will be specific challenges in each level that must be completed in order to advance in the game.

- **AI NPCs:**
  - The actions and strategies employed by the player will cause the NPC behavior to alter dynamically.

- **Cinematic Cutscenes:**
  - Immersive, narrative-driven cutscenes shall be incorporated into the system to set up the plot and increase player involvement.
  - At crucial points in the story, cutscenes will be incorporated into the gaming system.

- **Dynamic Sound and Lighting:**
  - The game will have a 3D positional audio system that changes its sound according to the player's movements and location.
  - In response to in-game events, the lighting will change to create an immersive atmosphere.

- **Mini-Map:**
  - The system will provide a mini-map that displays the player's current location along with important locations such as safe zones and objectives.
  - The mini-map will be updated in real time while the player investigates their surroundings.

3.3 Quality Attributes

Performance:

- To avoid motion sickness in virtual reality settings and to guarantee fluid gameplay, the game should maintain a minimum frame rate of 60.

- For the sake of the player experience, shorter load times are preferable, particularly when loading levels or assets.

Accessibility:

- User Interface should be simple to use so that player can easily use it

Immersion:

- Realism in visuals, audio, and lighting should all contribute to a highly immersive environment and improve the player's experience.
- Narrative components and cinematic cut-scenes should captivate players and strengthen their sense of immersion in the game world.

Reliability:

- For players to have a flawless gaming experience, the game should be stable with few crashes or glitches.
- If the save feature is reliable, players should be able to pick up where they left off without losing any progress.

Security:

- User data must be transmitted and stored securely to avoid illegal access and security lapses, particularly when it comes to player progress and login credentials.

Scalability:

- Future game modes or level additions should be supported by the game architecture without requiring extensive reworking.

Maintenance:

- The codebase should be well-documented and modular to facilitate future updates and bug fixes.
- Frequent software updates are necessary to ensure compatibility with new VR hardware, fix bugs, and introduce new features.

Interaction:

- Players should dynamically interact with the environment and game mechanics, providing a highly interactive experience.
- AI NPCs should behave realistically and adapt to the player's actions to provide a fair and challenging experience.

Audio and Visual Quality:

- High-quality graphics and textures should be used to create a visually appealing game environment that enhances immersion.
- Dynamic lighting and sound effects must respond to player actions, contributing to the overall atmosphere and tension in the game.

3.4 Non-Functional Requirements

Performance:

- The game must maintain a minimum frame rate of 60 FPS on supported VR hardware.
- Loading times between levels should not exceed 5 seconds.
- User inputs (e.g., hand tracking, VR controllers) should be processed with a maximum latency of 100 milliseconds to ensure a responsive experience.

Usability:

- All in-game menus must be navigable within two steps from the main menu.

Security:

- User data, including passwords, must be encrypted using industry standards.
- Firebase Authentication with encrypted communication (SSL/TLS) must be used for user authentication.

Reliability:

- The system should have a 99% uptime, excluding scheduled maintenance.
- The game should automatically save progress every 5 minutes during gameplay to prevent loss.

Scalability and Maintainability:

- The architecture should support the addition of new features, levels, or modes without requiring major reworking (e.g., modular design).

Compatibility:

- The game must be compatible with Oculus Quest and other supported VR platforms.
- The game must support VR controllers with hand tracking and haptic feedback.

Compliance:

- The game must comply with age-rating guidelines based on content.
- All data handling and storage must comply with GDPR and other relevant data privacy regulations.

Before playing, users must agree to the game's consent policy, which states:

- The user certifies that they are 12 years of age or older.
- The user acknowledges that they do not have motion sickness or any condition that could be aggravated by the immersive nature of the VR horror experience.
- The user understands that any negative consequences of the immersive nature of the game, including any discomfort—either physical or emotional—are entirely their responsibility.

Chapter 4

Use Case Diagram

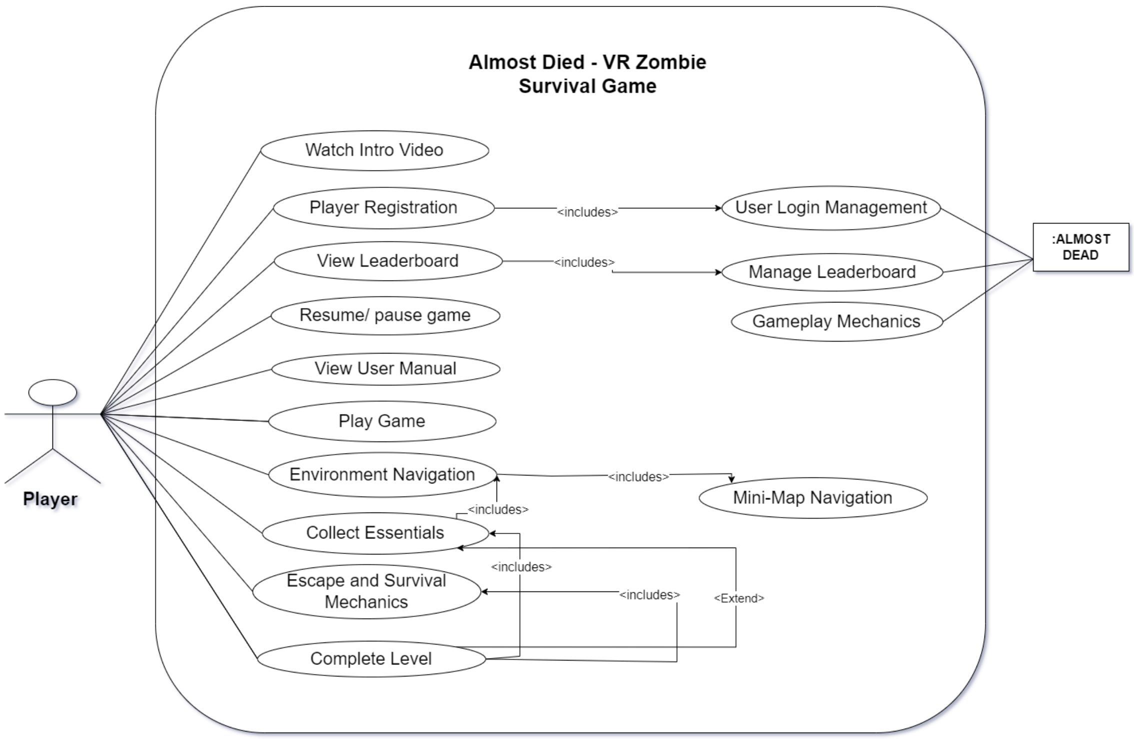

Figure 4.1: UseCase Diagram

Chapter 5

High-Level Use Cases

5.1 Player Registration

Use Case	Player Registration
Actors	Player
Type	Primary
Description	The player creates a new account by entering their email address, password, and username.

Table 5.1: Player Registration Use Case

5.2 User Login Management

Use Case	User Login Management
Actors	Player
Type	Primary
Description	Securely verifies players' login credentials using Firebase to manage player authentication.

Table 5.2: User Login Management Use Case

5.3 Watch Intro Video

Use Case	Watch Intro Video
Actors	Player
Type	Primary
Description	The opening video that establishes the backstory of the game is viewed by the player.

Table 5.3: Watch Intro Video Use Case

5.4 View Leaderboard

Use Case	View Leaderboard
Actors	Player
Type	Primary
Description	Shows player rankings according to level completion, performance, and survival time.

Table 5.4: View Leaderboard Use Case

5.5 Resume/Pause Game

Use Case	Resume/Pause Game
Actors	Player
Type	Primary
Description	Enables players to put the game on hold and pick it back up while playing.

Table 5.5: Resume/Pause Game Use Case

5.6 View User Manual

Use Case	View User Manual
Actors	Player
Type	Primary
Description	For help and direction, the player can consult the in-game manual.

Table 5.6: View User Manual Use Case

5.7 Play Game

Use Case	Play Game
Actors	Player
Type	Primary
Description	The primary gameplay involves players navigating the game world and trying to survive.

Table 5.7: Play Game Use Case

5.8 Mini-Map Navigation

Use Case	Mini-Map Navigation
Actors	Player
Type	Primary
Description	A mini-map helps guide and assist the player in navigating the environment to accomplish goals.

Table 5.8: Mini-Map Navigation Use Case

5.9 Environment Navigation

Use Case	Environment Navigation
Actors	Player
Type	Primary
Description	Enables players to navigate and explore the levels and environments within the game.

Table 5.9: Environment Navigation Use Case

5.10 Collect Essentials

Use Case	Collect Essentials
Actors	Player
Type	Primary
Description	In order to survive and advance through the levels, the player must gather essentials like food and tools.

Table 5.10: Collect Essentials Use Case

5.11 Escape and Survival Mechanics

Use Case	Escape and Survival Mechanics
Actors	Player
Type	Primary
Description	The player engages in non-violent survival mechanics such as puzzle-solving and avoiding danger.

Table 5.11: Escape and Survival Mechanics Use Case

5.12 Complete Level

Use Case	Complete Level
Actors	Player
Type	Primary
Description	The player finishes the level when they have met all the requirements and are still alive.

Table 5.12: Complete Level Use Case

5.13 Manage Leaderboard

Use Case	Manage Leaderboard
Actors	Game System
Type	Secondary
Description	The player leaderboard is maintained and updated with new scores by the game system.

Table 5.13: Manage Leaderboard Use Case

5.14 Gameplay Mechanics

Use Case	Gameplay Mechanics
Actors	Game System
Type	Secondary
Description	Manages the fundamental logic of the game, including player movement, interaction, and survival components.

Table 5.14: Gameplay Mechanics Use Case

Chapter 6

Extended Use Case

6.1 Player Registration

Use Case	Player Registration
Actors	Player
Scope	Almost Dead
Level	User Goal
Stakeholders and Interests	**Player**: In order to save the progress and login, the user wants to create the secure account **Game Developer**: Make sure players data is protected and accounts are created securely.
Preconditions	The game is installed on the player's device, and they are online. Firebase Authentication is integrated into the game system.
Success Guarantee (Post Conditions)	The player's account is successfully created and safely stored. A confirmation email is sent to the player's registered email.

<table>
<tr><td rowspan="1">Main Success Scenario</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. The player selects "Register" from the main menu</td><td>2. The system prompts the player to enter a username, email, and password.</td></tr>
<tr><td></td><td>3. The system verifies the information.</td></tr>
<tr><td></td><td>4. The account is created using Firebase Authentication.</td></tr>
<tr><td>5. The player is logged in automatically, and their progress is saved.</td><td></td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Already Registered Email: The system detects when the player’s email address is already in use. The player is asked to use a new email address.
Email Invalid: The system asks the user to enter an valid email format.
Weak Password: The system ask the user to create a stronger password.</td></tr>
<tr><td>Special Requirements</td><td>The registration process must be completed within 10 seconds.
Compliance with GDPR for secure handling of user data.</td></tr>
</table>

6.2 User Login Management

Use Case	User Login Management
Actors	Player
Scope	Almost Dead
Level	User Goal
Stakeholders and Interests	**Player:** wants to securely login and get there previously saved game progress and data **Game Developers:** make sure that player data is securely accessed .
Preconditions	The player’s account has already been created. The Firebase Authentication system is operational.
Success Guarantee (Post Conditions)	The player has logged in and been verified. The player’s game progress is loaded.

<table>
<tr><td rowspan="4">Main Success Scenario</td><td>Actor</td><td>System</td></tr>
<tr><td>1. The player goes to the main menu and chooses "Login."</td><td>2. The player is prompted to enter their password and email address by the system.</td></tr>
<tr><td></td><td>3. By using Firebase Authentication, the system confirms the credentials.</td></tr>
<tr><td></td><td>4. The player’s progress is restored, and they are logged in.</td></tr>
<tr><td>Extensions</td><td colspan="2">Wrong Password: The system ask user to change the password by using forget password option
Forgotten Email: The system provides a means for the user to retrieve their account.</td></tr>
<tr><td>Special Requirements</td><td colspan="2">The login process ought to be finished within five seconds.</td></tr>
</table>

6.3 Watch Intro Video

Use Case	Watch Intro Video
Actors	Player
Scope	Almost Dead
Level	User Goal
Stakeholders and Interests	**Player**:In order to know about the plot of tha game player will watch intro video **Game developer:**To increase player interest in game, make sure the video should be engaging and immersive
Preconditions	The video is available, and the game has begun. The video has not been skipped by the user. .
Success Guarantee (Post Conditions)	The full introduction video is seen by the player. After the video, the game switches to the main menu.

<table>
<tr><td rowspan="4">Main Success Scenario</td><td>Actor</td><td>System</td></tr>
<tr><td></td><td>1. The opening video plays when the game launches.</td></tr>
<tr><td>2. The game’s backstory is established when the player watches the video.</td><td></td></tr>
<tr><td></td><td>3. The video ends, and the game opens to the main menu.</td></tr>
<tr><td>Extensions</td><td colspan="2">Skip Video: The game jumps straight to the main menu when the player chooses to skip the video.</td></tr>
<tr><td>Special Requirements</td><td colspan="2">The video needs to be rendered in high definition and load within 10 seconds or less.</td></tr>
</table>

6.4 View Leaderboard

Use Case	View Leaderboard
Actors	Player
Scope	Almost Dead
Level	User Goal
Stakeholders and Interests	**Player**: Has to check their ranking in comparison to other players. **Game developer:** makes sure that the leaderboard is updated in real time.
Preconditions	The player has been authenticated. The leaderboard data is current and readily available. .
Success Guarantee (Post Conditions)	A leaderboard with precise rankings is displayed to the player.

Main Success Scenario	**Actor**	**System**
	1. The player opens the main menu and chooses "View Leaderboard."	
		2. The system gets the most recent leaderboard information.
		3. Player rankings based on performance, survival time, and level completion are shown on the leaderboard.
Extensions	**No Internet Connection** : A notification of the connection problem is sent to the player, asking them to try again.	
Special Requirements	After a rank change, the leaderboard should be updated in two seconds.	

6.5 Resume/Pause Game

Use Case	Resume/Pause Game
Actors	Player
Scope	Almost Dead
Level	User Goal
Stakeholders and Interests	**Player**: Would prefer to pause when needed and start again from where they left. **Game developer:**ensure that the game state is accurately saved and restored with 1 sec lag.
Preconditions	The game is currently being played. .
Success Guarantee (Post Conditions)	The game is paused, and the player picks up where they left off.

<table>
<tr><td>Main Success Scenario</td><td><table><tr><th>Actor</th><th>System</th></tr><tr><td>1. The player opens the main menu and chooses "View Leaderboard."</td><td></td></tr><tr><td></td><td>2. The system gets the most recent leaderboard information.</td></tr><tr><td></td><td>3. Player rankings based on performance, survival time, and level completion are shown on the leaderboard.</td></tr></table></td></tr>
<tr><td>Extensions</td><td>Timeout or Disconnection: The player picks up where they left off at the last checkpoint when the game automatically saves.</td></tr>
<tr><td>Special Requirements</td><td>There should be a maximum one-second lag between the pause and resume functions.</td></tr>
</table>

6.6 View User Manual

Use Case	View User Manual
Actors	Player
Scope	Almost Dead
Level	User Goal
Stakeholders and Interests	**Player**: Would like assistance and direction from the in-game manual. **Game developer:** makes sure the instruction manual is clear and easy to read.
Preconditions	The in-game manual is available to the player who is currently logged in. .
Success Guarantee (Post Conditions)	The guide can be successfully viewed by the player.

<table>
<tr><td>Main Success Scenario</td><td><table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. The user chooses "View User Manual" from the menu of options.</td><td></td></tr>
<tr><td></td><td>2. The user manual is shown by the system.</td></tr>
<tr><td>3. The player looks for instructions by scrolling through the handbook</td><td></td></tr>
</table></td></tr>
<tr><td>Extensions</td><td>Access During Play: Without changing the game’s state, the player can consult the manual at any time during play.</td></tr>
<tr><td>Special Requirements</td><td>The manual should be easily navigable.</td></tr>
</table>

6.7 Play Game

<table>
<tr><td>Use Case</td><td>Play Game</td></tr>
<tr><td>Actors</td><td>Player</td></tr>
<tr><td>Scope</td><td>Almost Dead</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Stakeholders and Interests</td><td>Player: make sure to use strategic decision making in order to escape and survive
Game developer: ensures that the game runs efficiently without any lag while keeping balance between difficulty and immersion.</td></tr>
<tr><td>Preconditions</td><td>After logging in, the player chose to "Play." .</td></tr>
<tr><td>Success Guarantee (Post Conditions)</td><td>The game world is successfully navigated by the player as they advance through the levels.</td></tr>
<tr><td>Main Success Scenario</td><td><table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. Within the main menu, the player chooses "Play Game."</td><td></td></tr>
<tr><td></td><td>2. System started the game</td></tr>
<tr><td></td><td>3. The game is resumed by the player after it has been paused.</td></tr>
</table></td></tr>
</table>

Extensions	**Player Death:** When a player passes away, the game lets them resume from a saved state.
Special Requirements	For seamless gameplay, the game needs to load in 10 seconds or less and run at 60 frames per second or higher.

6.8 Mini-Map Navigation

<table>
<tr><td>Use Case</td><td>Mini-Map Navigation</td></tr>
<tr><td>Actors</td><td>Player</td></tr>
<tr><td>Scope</td><td>Almost Dead</td></tr>
<tr><td>Level</td><td>Sub-Function</td></tr>
<tr><td>Stakeholders and Interests</td><td>Player: want to see the all the area and objects
Game developer: guarantees that the mini-map is updated in real time and provides gamers with clear and helpful information.</td></tr>
<tr><td>Preconditions</td><td>The player must keep track of targets or locations.
In-game, the player can access the mini-map.</td></tr>
<tr><td>Success Guarantee (Post Conditions)</td><td>Using the mini-map, the player successfully navigates the surroundings.
There is accurate display of the locations and goals.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. In order to access the mini-map while playing, the player must press the mini-map</td><td></td></tr>
<tr><td></td><td>2. The system shows the current surroundings, along with the main locations and objectives.</td></tr>
<tr><td></td><td>3. As the player moves, the mini-map updates in real-time .</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Map is Unavailable: If the player is in an area that has not yet been explored, the system will tell them that the mini-map is not available.</td></tr>
</table>

Special Requirements	The mini-map needs to load quickly so that gameplay does not lag. Real-time updates and zoom capabilities must be supported.

6.9 Environment Navigation

Use Case	Environment Navigation
Actors	Player
Scope	Almost Dead
Level	User Goal
Stakeholders and Interests	**Player**: need to navigate and explore the game levels to uncover objectives and survive. **Game developer:** Ensures that exploration mechanics and transitions are seamless.
Preconditions	There are goals or areas for the player to investigate in the game .
Success Guarantee (Post Conditions)	The player navigates the environment with ease and without any problems. There is accessibility to all important areas and goals.
Main Success Scenario	**Actor** / **System**: Actor: 1. The player initiates movement within the surroundings. Actor: 2. To progress in the game, the player must solve puzzles, accomplish goals, or find important locations. System: 3. The player's progress is updated by the system in response to their exploration.
Extensions	**Blocked Path:** To proceed, the player must locate a different route or figure out a puzzle when they come across a blocked pa.

Main Success Scenario:

Actor	System
1. The player initiates movement within the surroundings.	
2. To progress in the game, the player must solve puzzles, accomplish goals, or find important locations.	
	3. The player's progress is updated by the system in response to their exploration.

Special Requirements	To prevent latency and preserve a steady framerate of at least 60 frames per second, the environment ought to load dynamically.

6.10 Collect Essentials

<table>
<tr><td>Use Case</td><td>Collect Essentials</td></tr>
<tr><td>Actors</td><td>Player</td></tr>
<tr><td>Scope</td><td>Almost Dead</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Stakeholders and Interests</td><td>Player: must need to collect survival and escape essentials according to need in order to advance in the game
Game developer: ensure that the essentials are hide appropriately and are simple to collect.</td></tr>
<tr><td>Preconditions</td><td>There are collectible items in the level the player is in.
Everything is set up and prepared for pickup. .</td></tr>
<tr><td>Success Guarantee (Post Conditions)</td><td>The player succeeds in gathering every necessary item
In the player’s inventory, the items they have collected are added.</td></tr>
<tr><td>Main Success Scenario</td><td><table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. The player comes into the scene armed with necessities (food, tools, etc.).</td><td></td></tr>
<tr><td>2. To gather the item, the player must interact with it.</td><td></td></tr>
<tr><td></td><td>3. The item is placed in the player’s inventory, and the item collection is monitored by the system</td></tr>
</table></td></tr>
<tr><td>Extensions</td><td>Full Inventory: The player must choose which items to leave after being notified by the system that their inventory is full.</td></tr>
<tr><td>Special Requirements</td><td>There should be a maximum one-second lag between the pause and resume functions.</td></tr>
</table>

6.11 Escape and Survival Mechanics

<table>
<tr><td>Use Case</td><td>Escape and Survival Mechanics</td></tr>
<tr><td>Actors</td><td>Player</td></tr>
<tr><td>Scope</td><td>Almost Dead</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Stakeholders and Interests</td><td>Player: Wants to get out of the game and survive by using non-violent means.
Game developer: makes sure that the survival and escape mechanics are interesting and well-balanced.</td></tr>
<tr><td>Preconditions</td><td>The player is actively avoiding dangers and navigating the surroundings. The system has offered chances for escape or obstacles for survival. .</td></tr>
<tr><td>Success Guarantee (Post Conditions)</td><td>By successfully avoiding danger, the player completes the level. The player gets away to the next area of safety or leve</td></tr>
<tr><td>Main Success Scenario</td><td><table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. The player comes across a danger, like zombies or energy drainage.</td><td></td></tr>
<tr><td></td><td>2. To avoid or escape the threat, the player employs a combination of strategic decision-making and puzzle-solving techniques.</td></tr>
<tr><td></td><td>3. After surviving the encounter, the player advances to the following area or level.</td></tr>
</table></td></tr>
<tr><td>Extensions</td><td>Challenge Not Solved:The system provides a hint or retries a challenge if the player is unable to solve it.</td></tr>
<tr><td>Special Requirements</td><td>Sustaining immersion during high-stress moments requires seamless, lag-free performance.</td></tr>
</table>

6.12 Complete Level

<table>
<tr><td>Use Case</td><td>Complete Level</td></tr>
<tr><td>Actors</td><td>Player</td></tr>
<tr><td>Scope</td><td>Almost Dead</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Stakeholders and Interests</td><td>Player: After accomplishing every goal, they want to end the level. Game developer: Ensures accurate goal tracking and seamless level transitions.</td></tr>
<tr><td>Preconditions</td><td>The player has finished all of the level’s prerequisite tasks.
.</td></tr>
<tr><td>Success Guarantee (Post Conditions)</td><td>After completing the current level successfully, the player advances to the next. The game keeps track of your progress and lets you access the next level.</td></tr>
<tr><td>Main Success Scenario</td><td><table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. The player accomplishes every level goal</td><td>2. The accomplishment of goals is confirmed by the system</td></tr>
<tr><td>3. To end the level, the player must either reach the exit or finish the last task</td><td>4.The main menu or the next level are navigated to by the system.</td></tr>
</table></td></tr>
<tr><td>Extensions</td><td>Incomplete Objectives: The player is prompted to finish the remaining tasks by the system if they attempt to finish the level without completing all of the objectives.</td></tr>
<tr><td>Special Requirements</td><td>Smooth level transitions should take no longer than five seconds to load.</td></tr>
</table>

6.13 Gameplay Mechanisms

Use Case	Gameplay Mechanisms
Actors	Player
Scope	Almost Dead
Level	Supporting Function

<table>
<tr><td>Stakeholders and Interests</td><td>Game Developer: is responsible for ensuring that essential gameplay elements—such as player movement, interaction, and survival—work flawlessly.</td></tr>
<tr><td>Preconditions</td><td>Gameplay mechanics are enabled, and the player is in the game.</td></tr>
<tr><td>Main Success Scenario</td><td><table>
<tr><th>Actor</th><th>System</th></tr>
<tr><td>1. The player uses survival mechanics, interacts with objects, and explores the game world.</td><td>2. The system processes player inputs and reflects them in real-time.</td></tr>
<tr><td>3. To end the level, the player must either reach the exit or finish the last task.</td><td>4. The system navigates to the main menu or the next level.</td></tr>
</table></td></tr>
<tr><td>Extensions</td><td>Incomplete Objectives: If the player attempts to finish the level without completing all the objectives, the system prompts them to finish the remaining tasks.</td></tr>
<tr><td>Special Requirements</td><td>The environment ought to load dynamically to prevent latency and preserve a steady frame rate of at least 60 frames per second to ensure smooth gameplay.</td></tr>
</table>

6.14 Manage Leaderboard

Use Case	Manage Leaderboard
Actors	Player
Scope	Almost Dead
Level	Supporting Function
Stakeholders and Interests	**Game developer:**makes sure that the scores are accurate and that the leaderboard is updated in real time.
Preconditions	For leaderboard ranking, player performance data is available. A level has been completed by the player.
Success Guarantee (Post Conditions)	The player's ranking and score are updated on the leaderboard. The leaderboard displays the latest results.

<table>
<tr><td>Main Success Scenario</td><td><table><tr><th>Actor</th><th>System</th></tr><tr><td>1.</td><td>1. After a level, the system determines how well the player performed.</td></tr><tr><td>3. To end the level, the player must either reach the exit or finish the last task</td><td>2.The new score is updated on the leaderboard by the system</td></tr></table></td></tr>
<tr><td>Extensions</td><td>Absence of Internet Connection: The player is notified by the system and the leaderboard does not update if there is no connection.</td></tr>
<tr><td>Special Requirements</td><td>Updates to the leaderboard have to happen quickly and without any lag.</td></tr>
</table>

Chapter 7

Domain Model

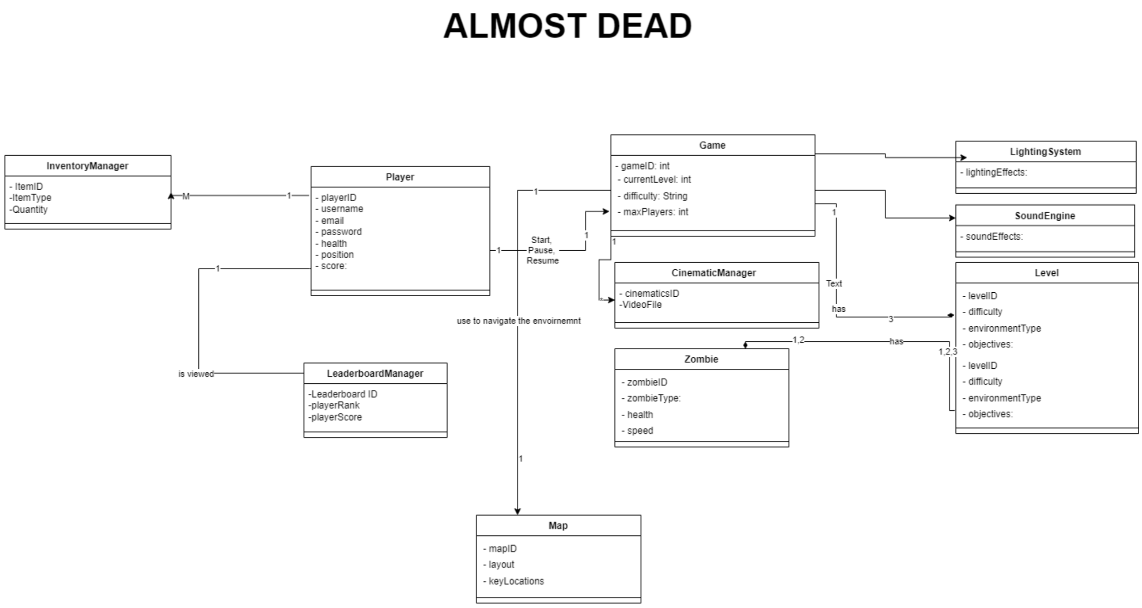

Figure 7.1: Domain Model Diagram

Chapter 8

System Sequence Diagram

8.1 Player Registration

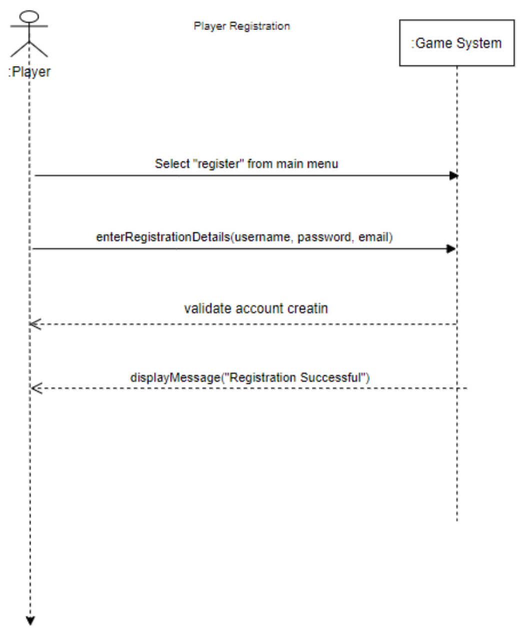

Figure 8.1: ssd Player Registration

8.2 User Login Management

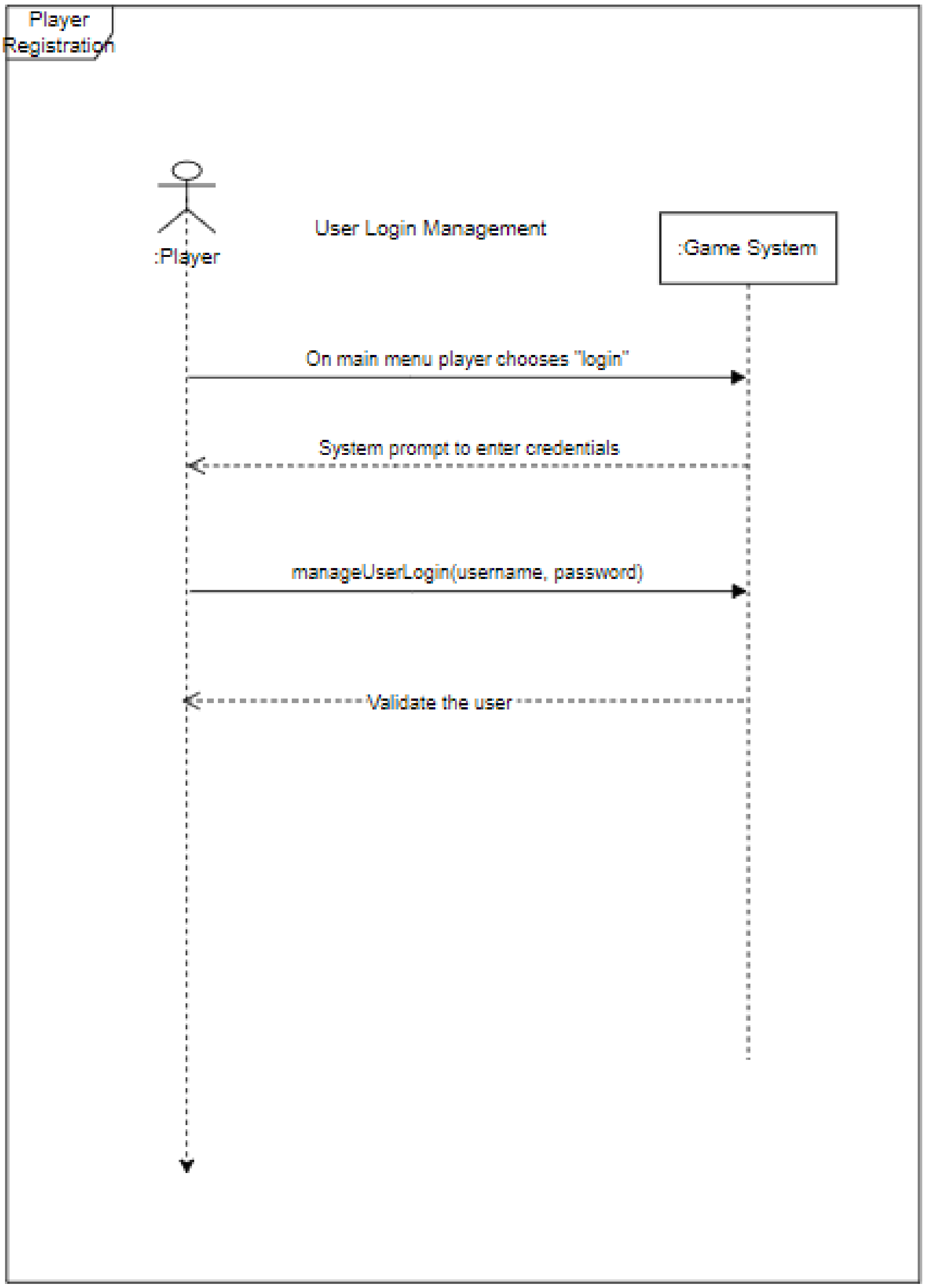

Figure 8.2: ssd User Login Managemnt

8.3 Watch Intro Video

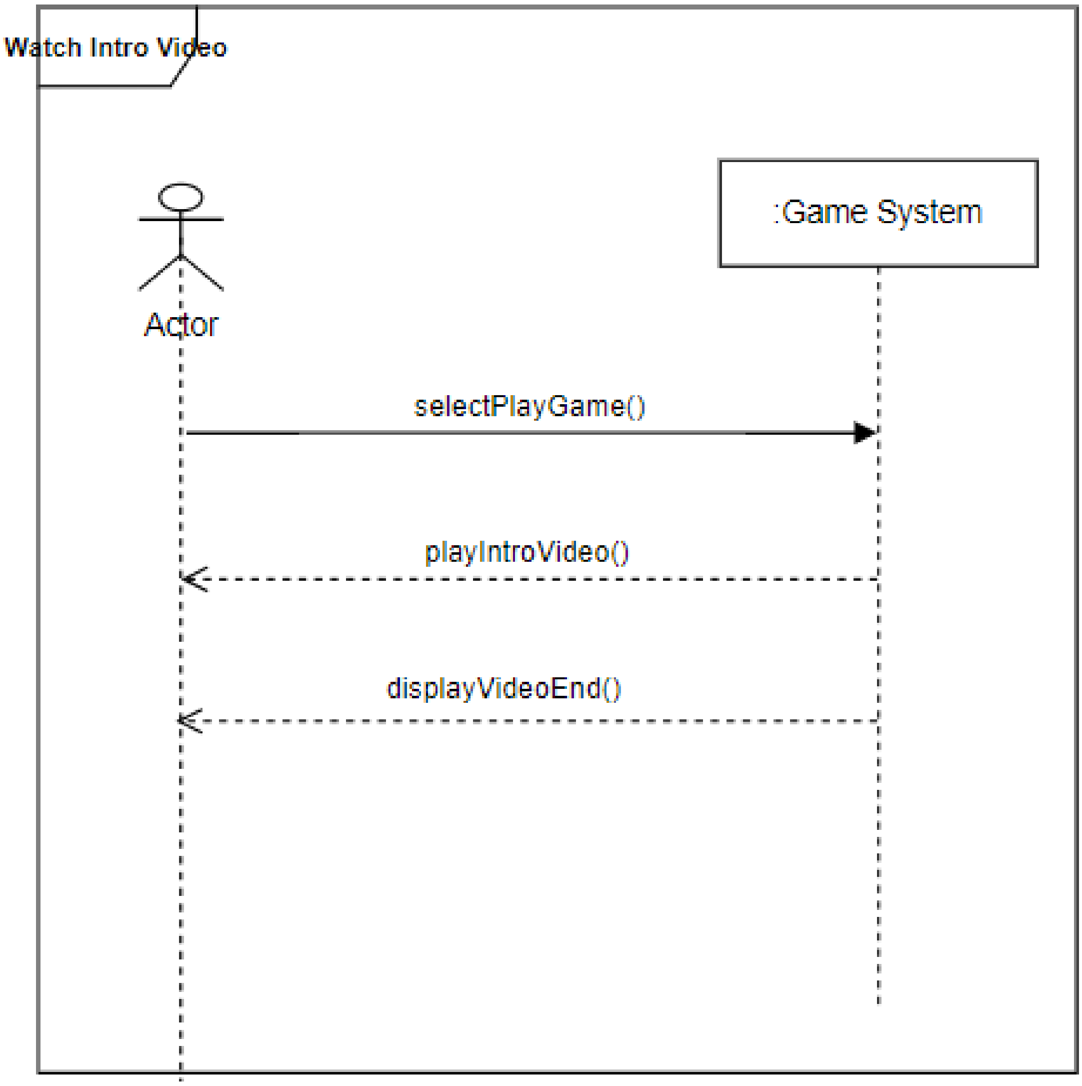

Figure 8.3: ssd Watch Intro Video

8.4 View Leader Board

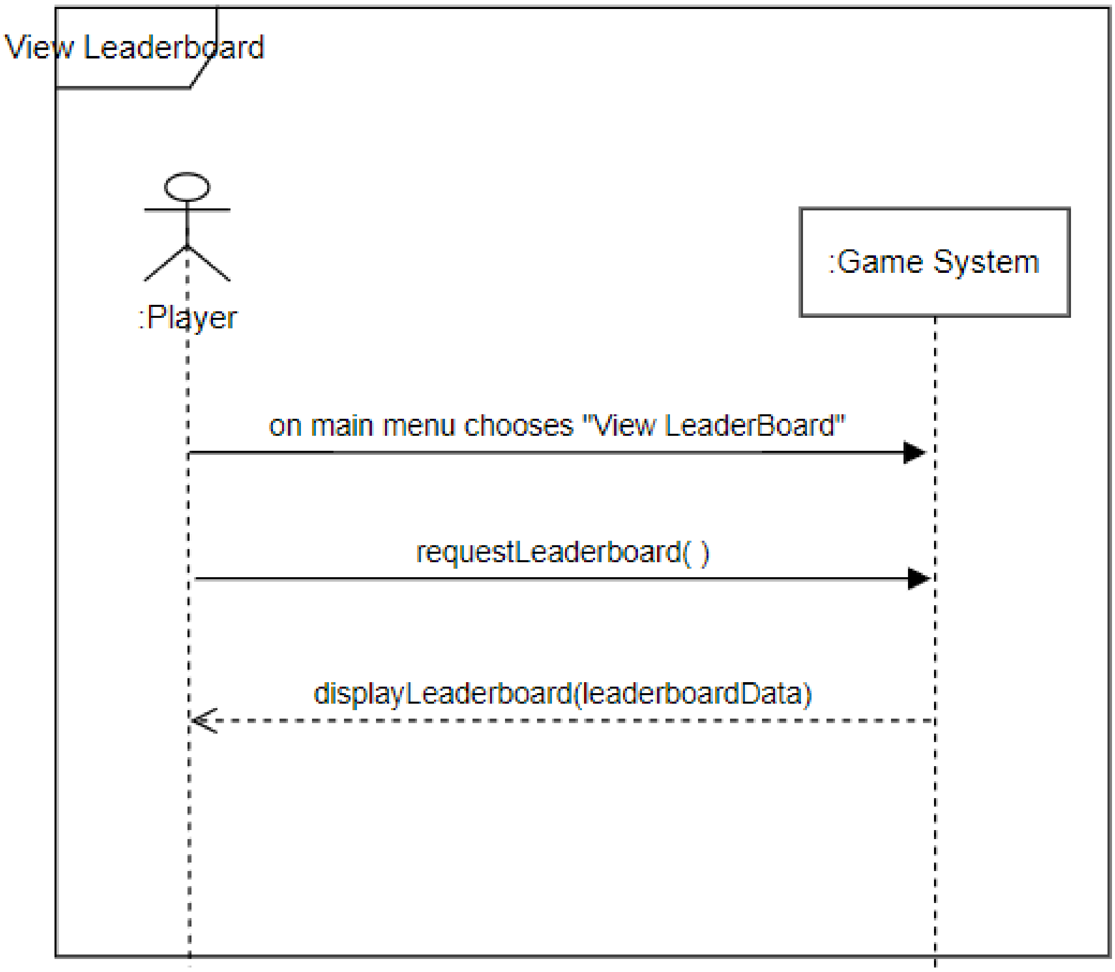

Figure 8.4: SSD View Leader BoarD

8.5 Resume/Pause Game

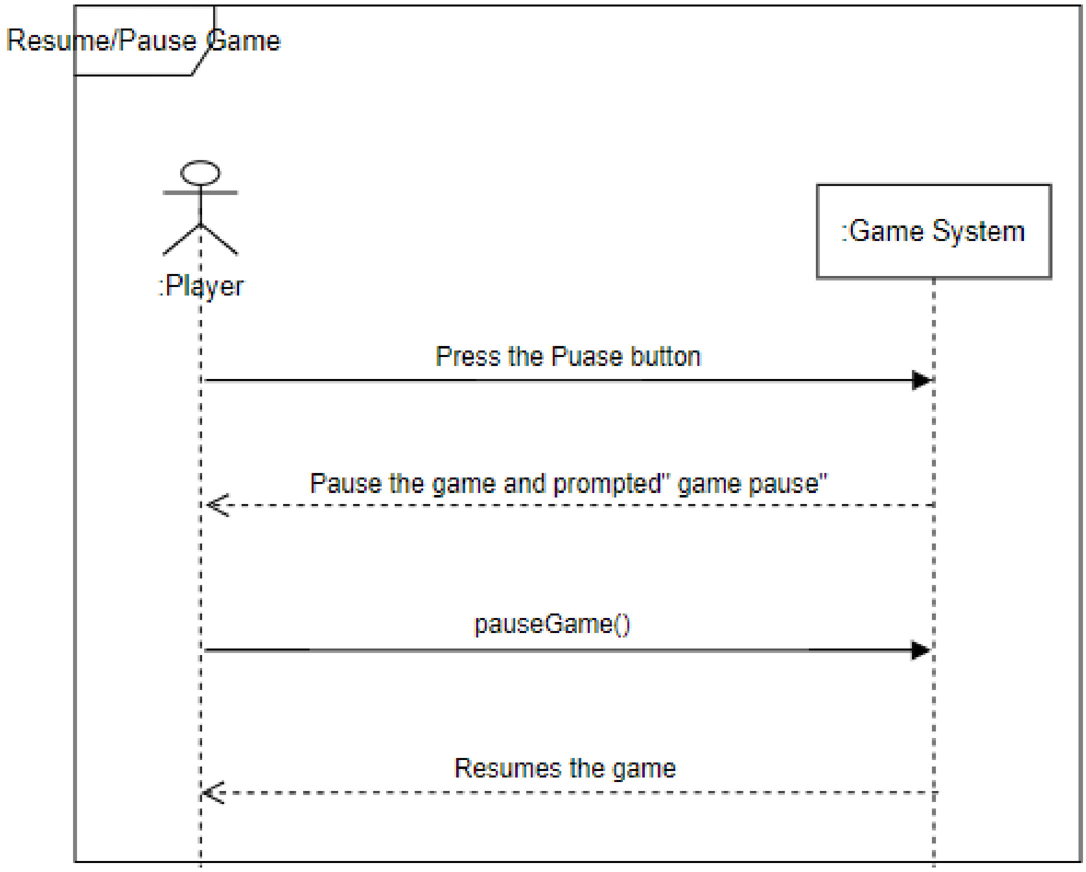

Figure 8.5: ssd Play/Resume Game

8.6 View User Manual

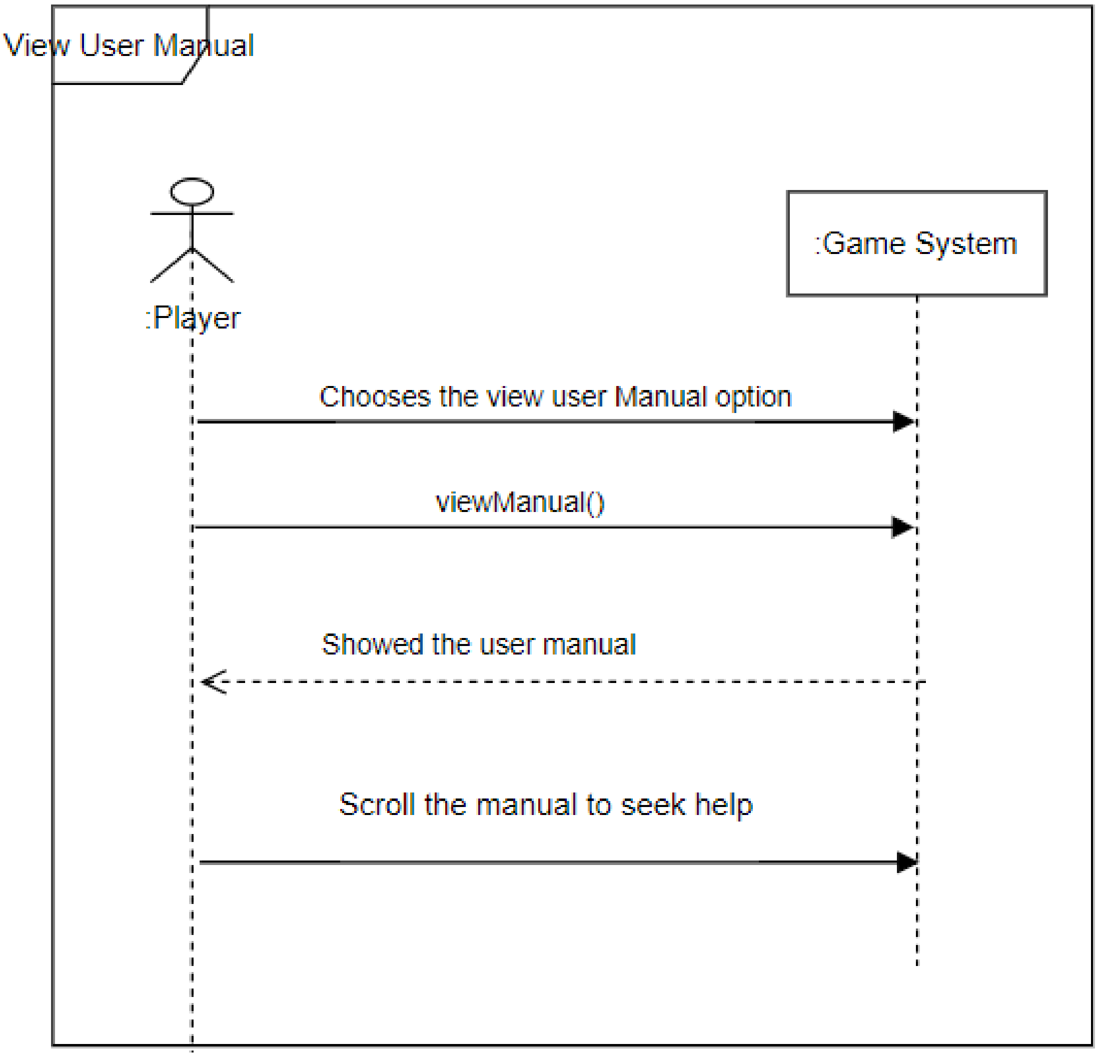

Figure 8.6: ssd View User Manual

8.7 Play Game

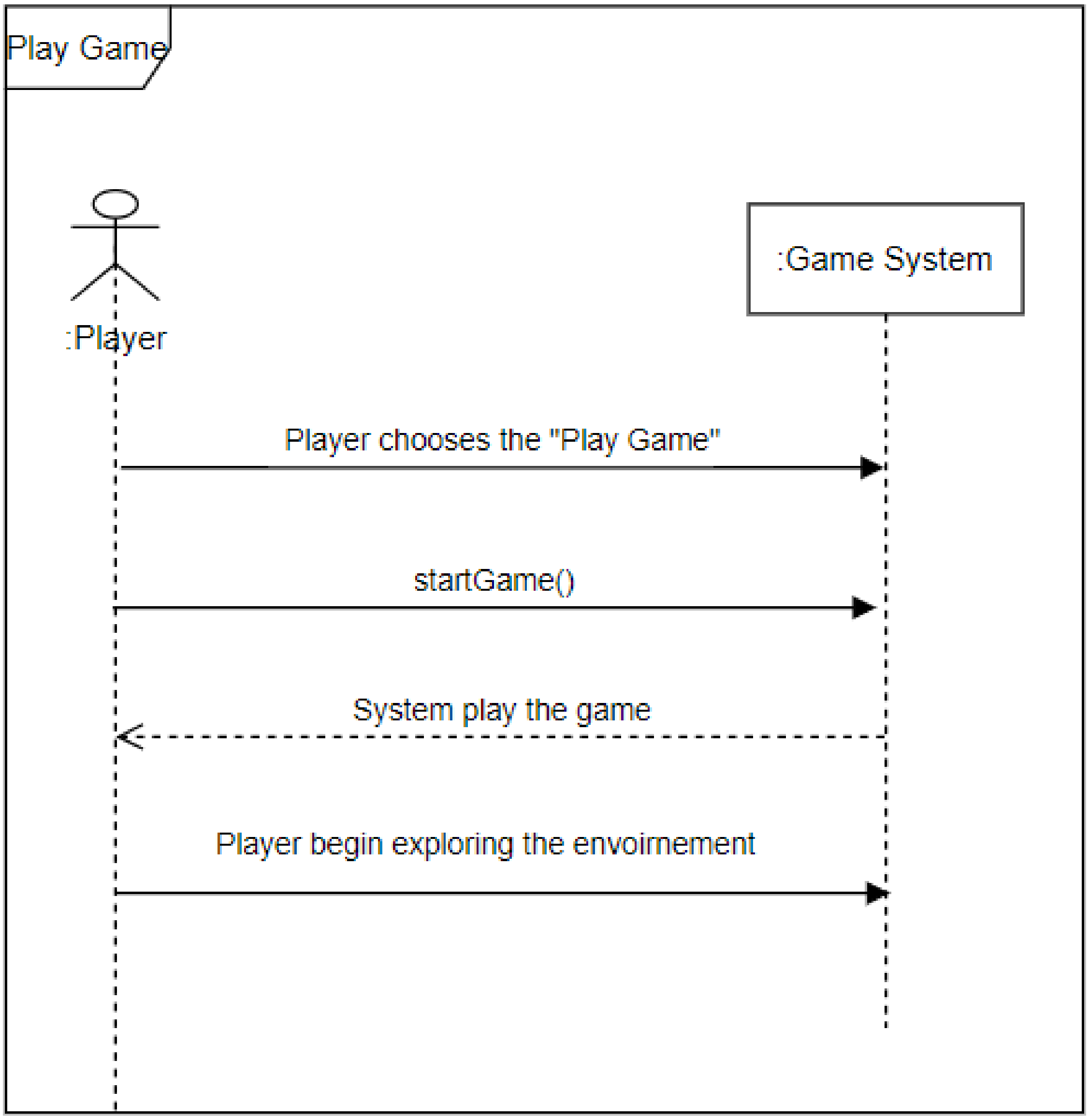

Figure 8.7: ssd Play Game

8.8 Mini-Map Navigation

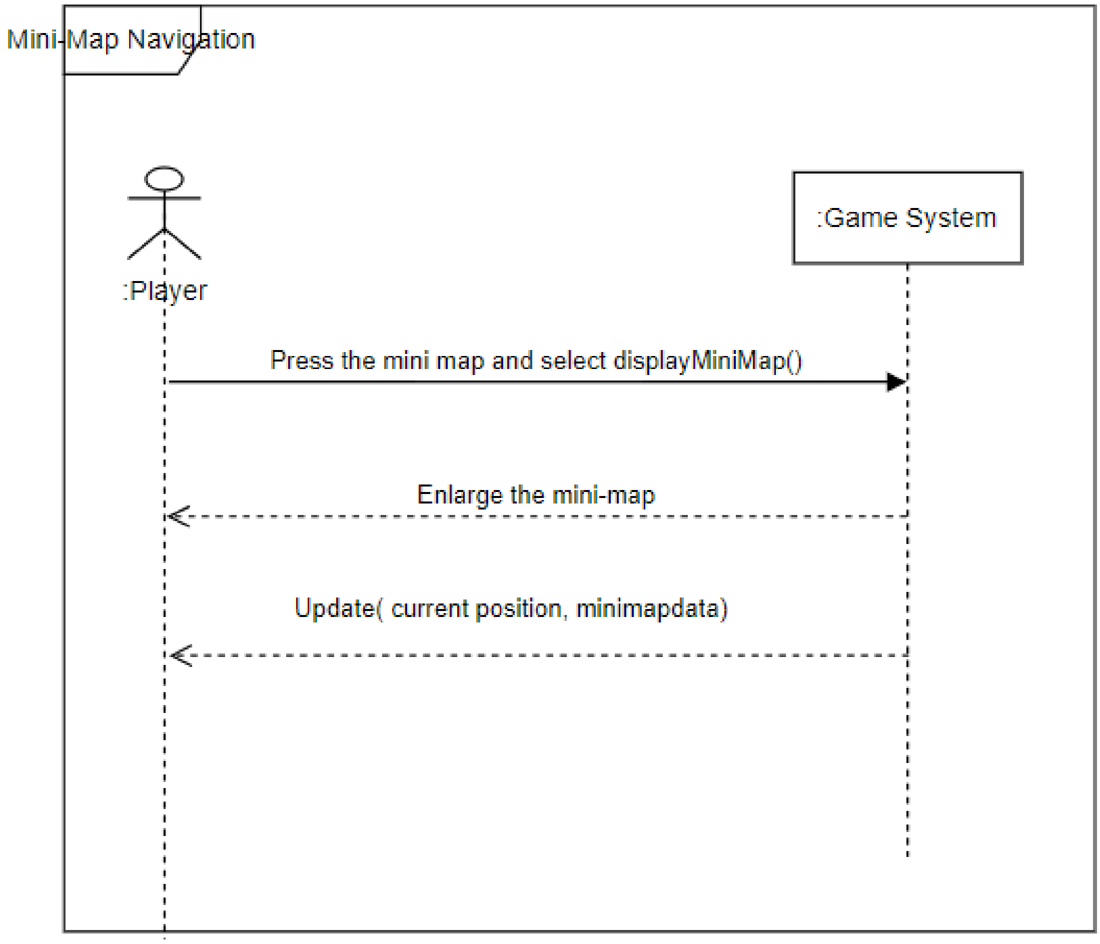

Figure 8.8: ssd Mini-Map Navigation

8.9 Environment Navigation

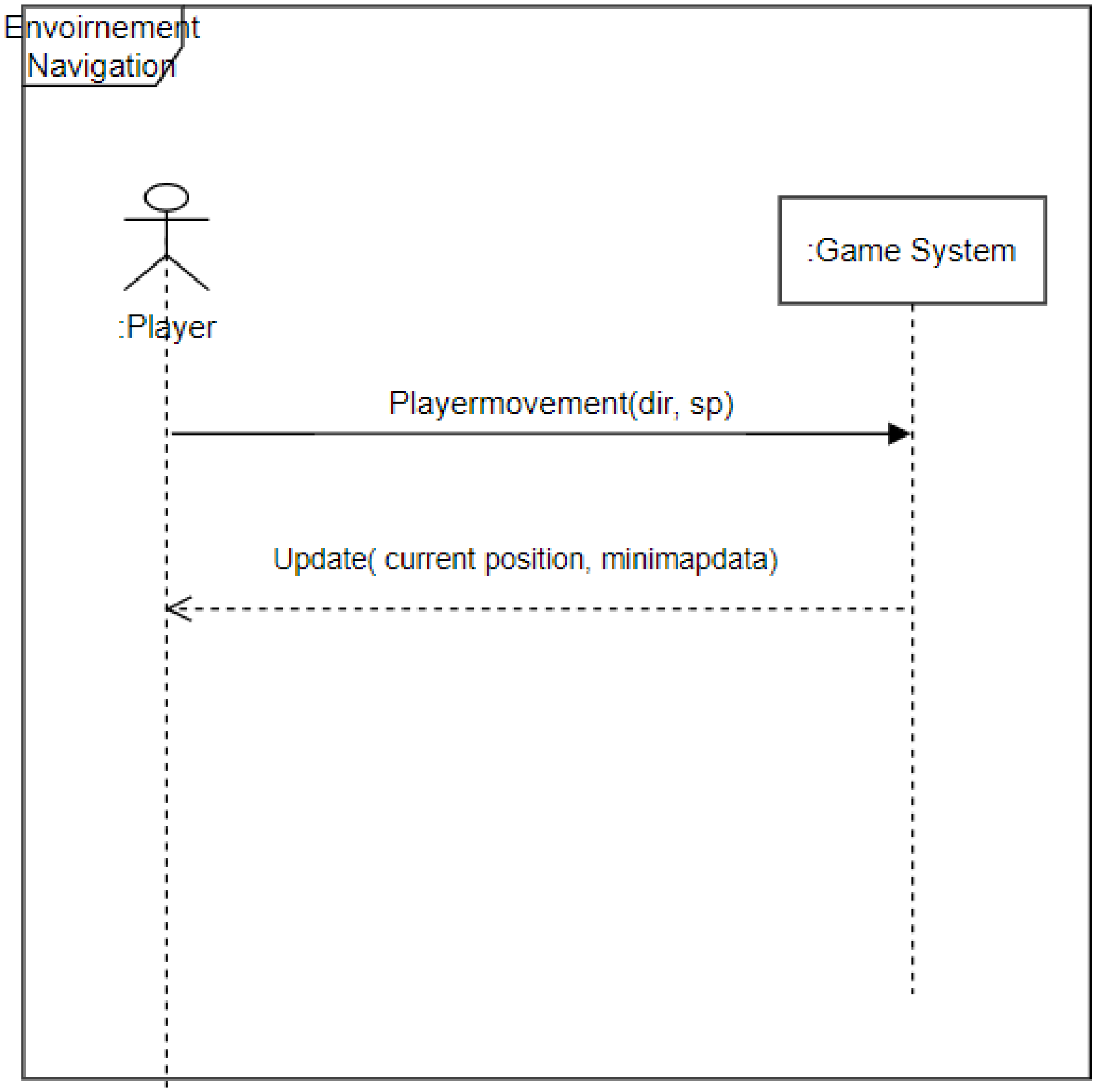

Figure 8.9: Environment Navigation

8.10 Collect Essential

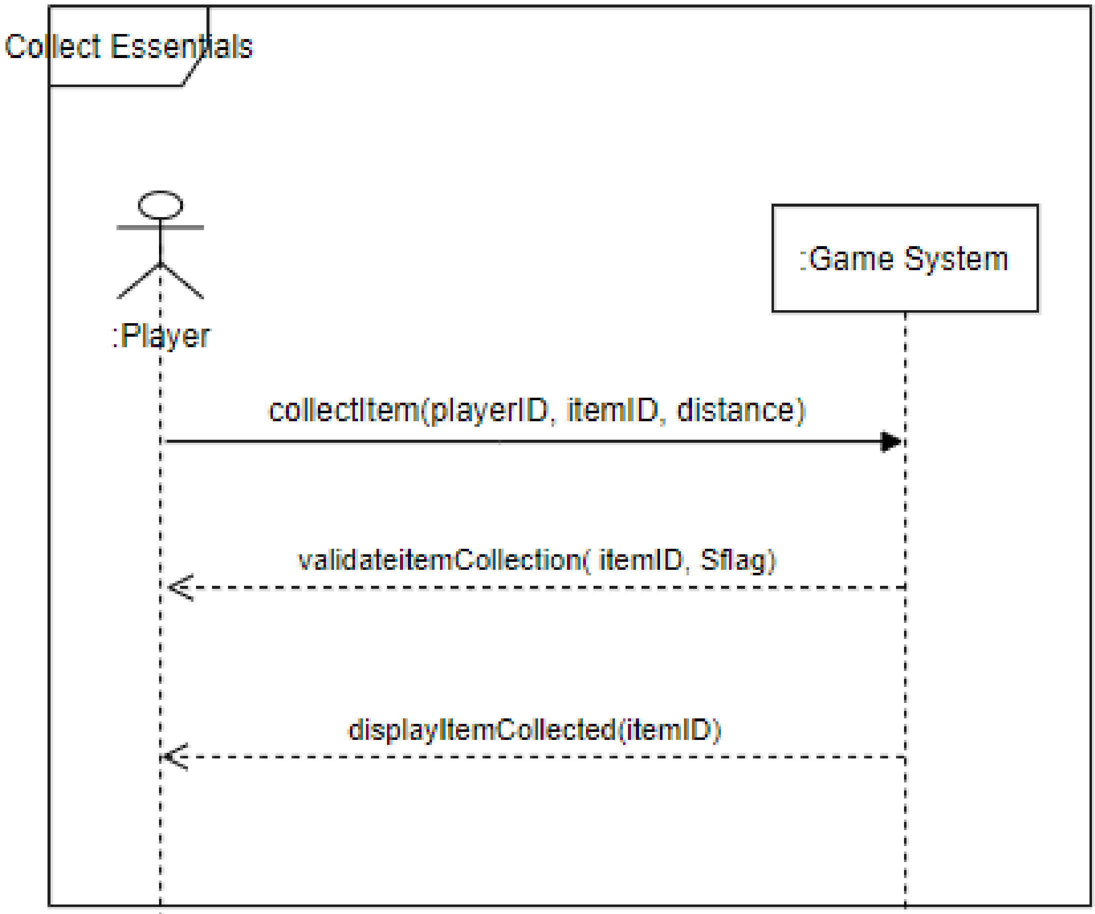

Figure 8.10: ssd Collect Essentials

8.11 Escape And Survival Mechanics

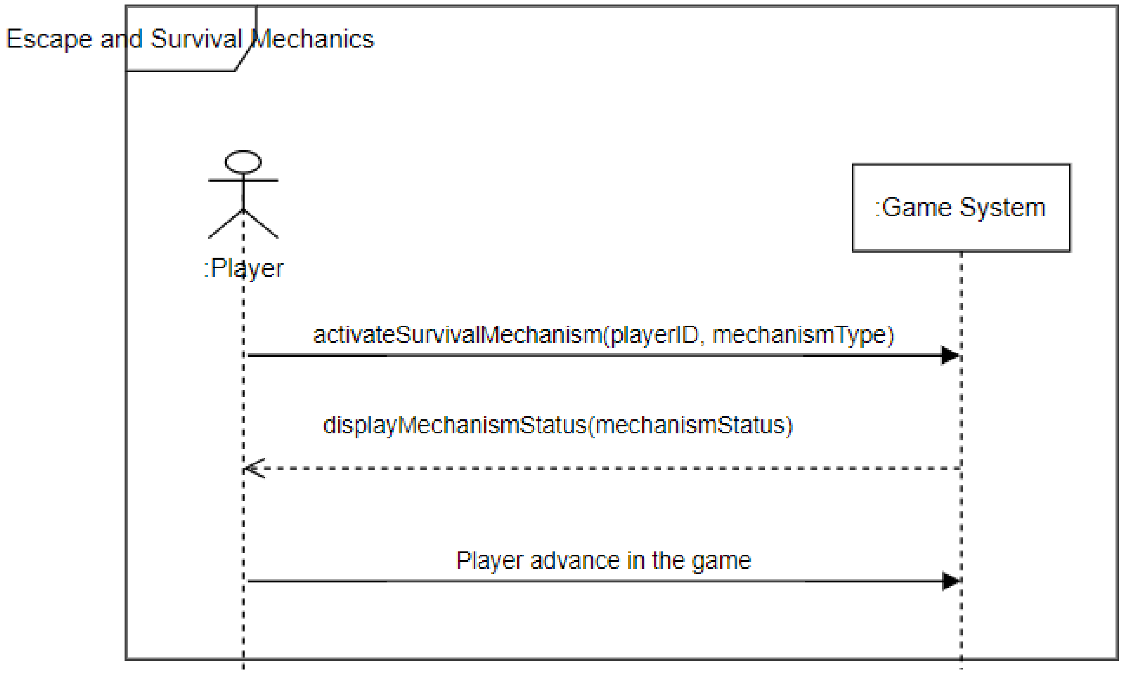

Figure 8.11: ssd Escape and survival Mechanics

8.12 Complete Level

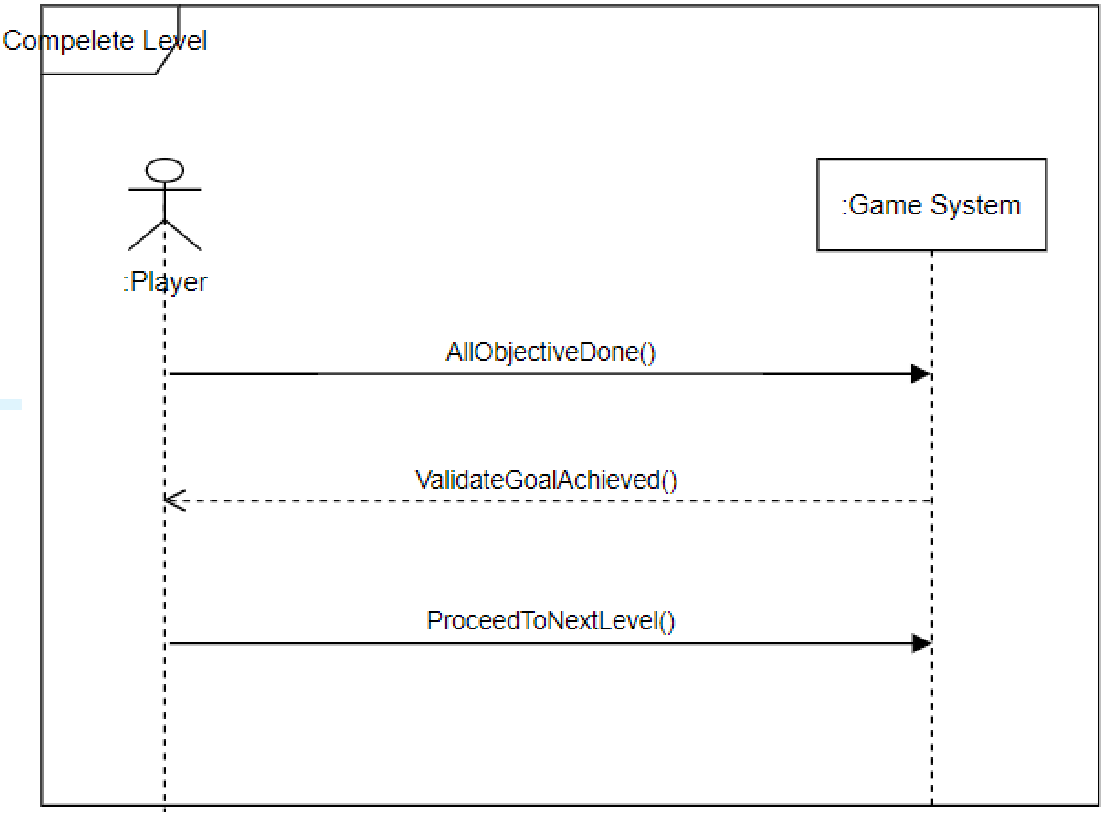

Figure 8.12: ssd Complete Level

8.13 Gameplay Mechanics

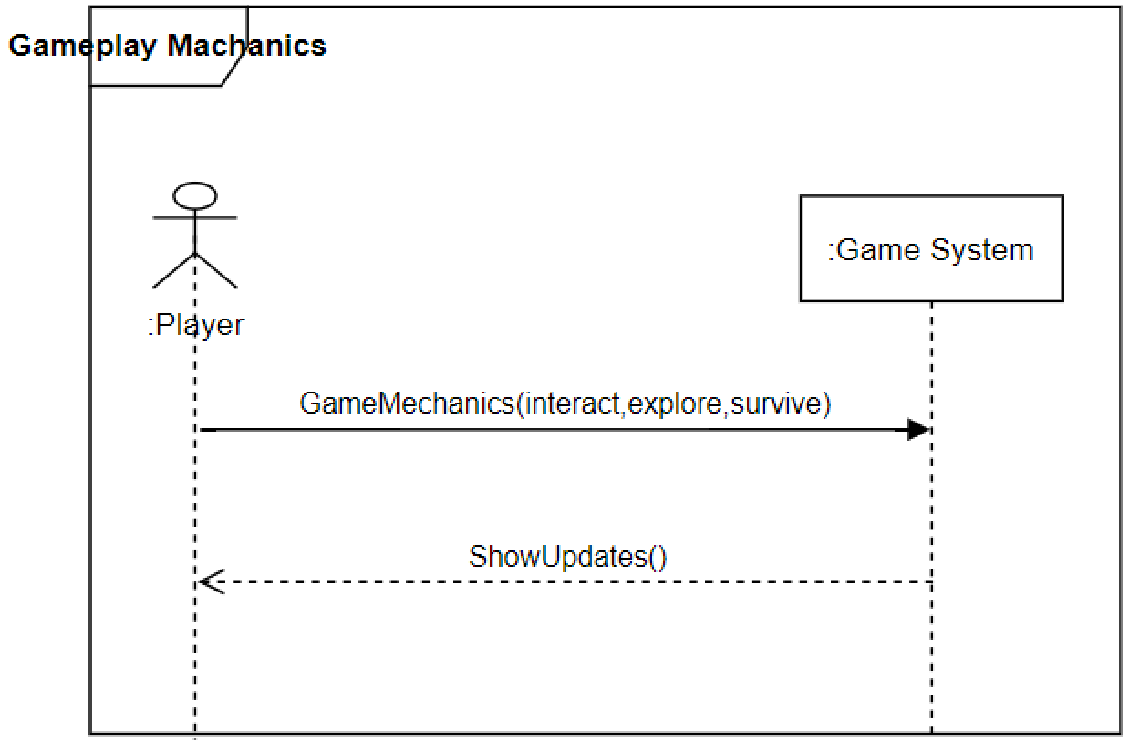

Figure 8.13: ssd Gameplay Mechanics

8.14 Manager Leader Board

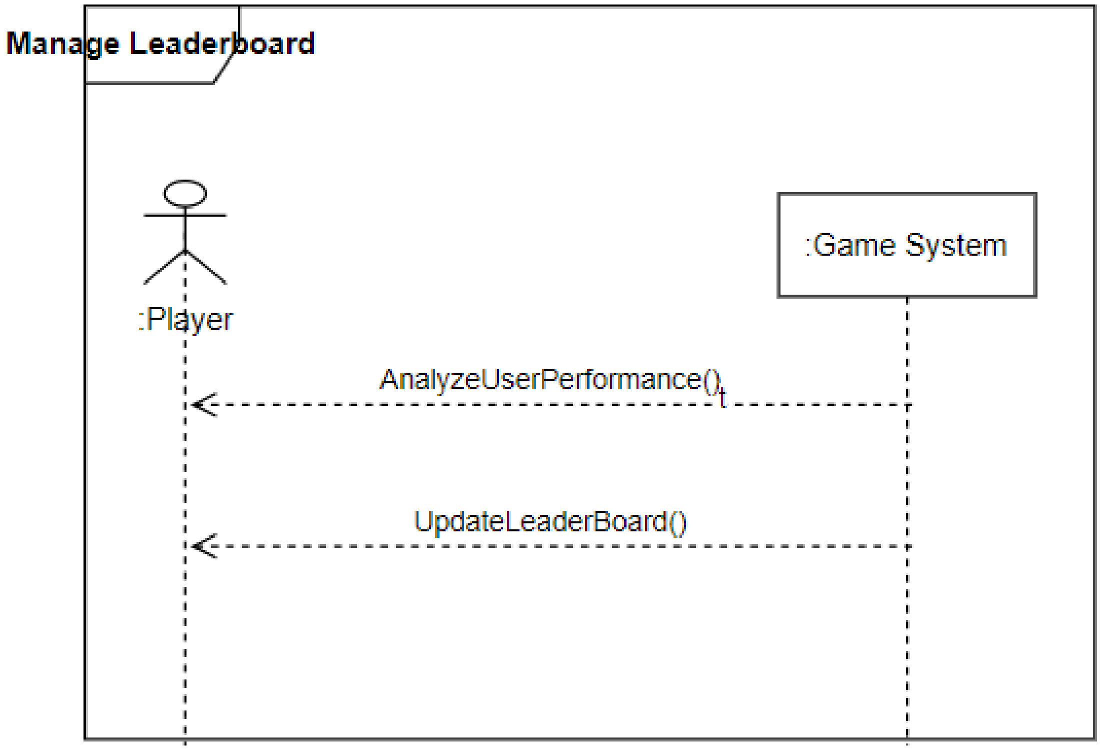

Figure 8.14: ssd Manage LeaderBoard

Chapter 9

Activity Diagram

9.1 Player Registration

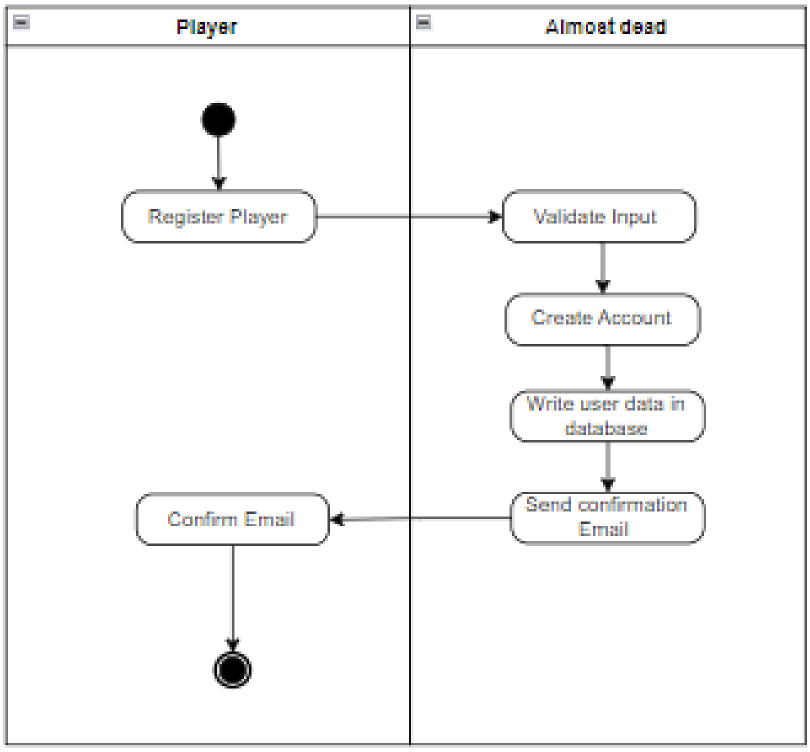

Figure 9.1: Enter Caption

9.2 User Login Management

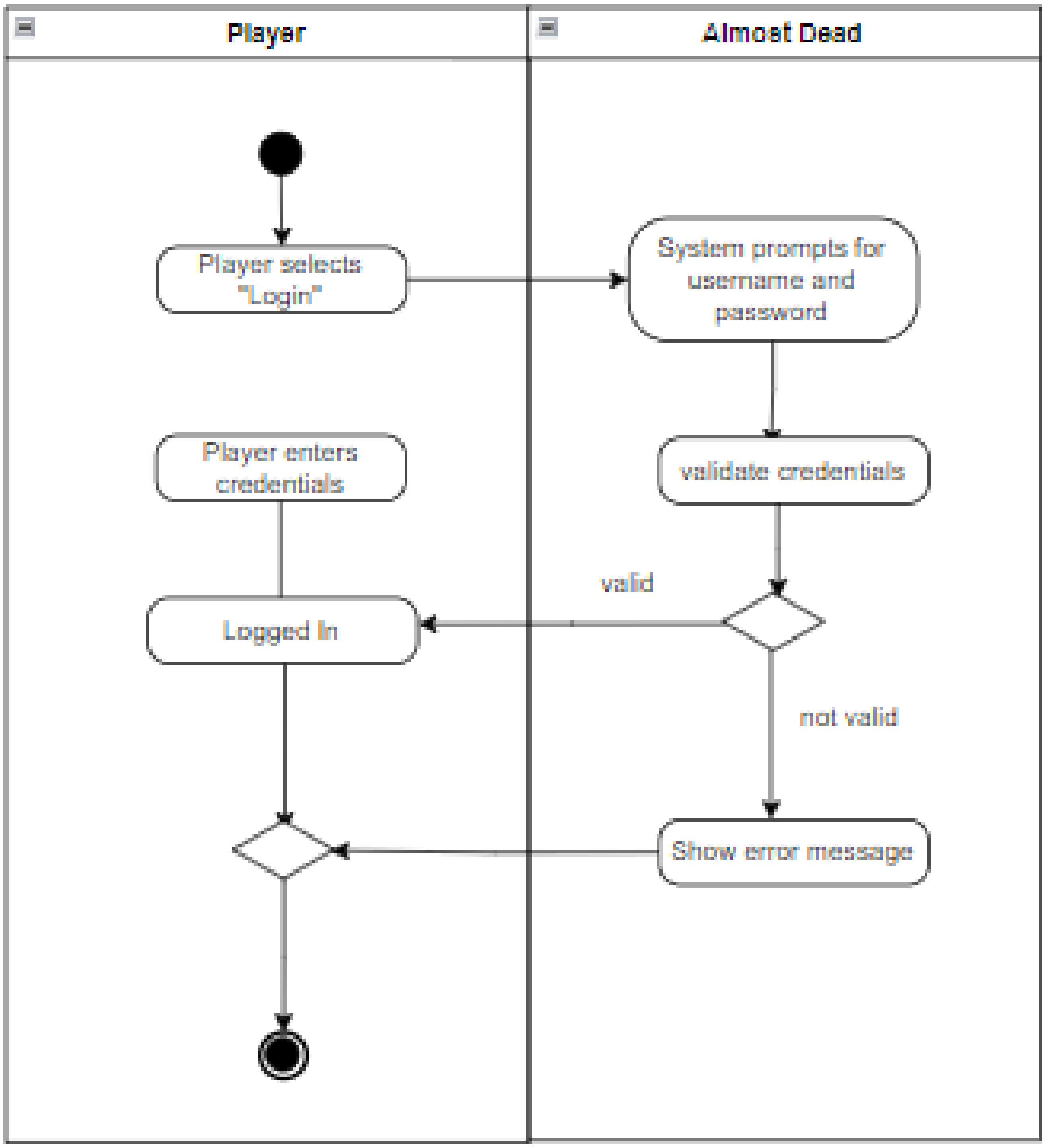

Figure 9.2: AD User Login Management

9.3 View Leader Board

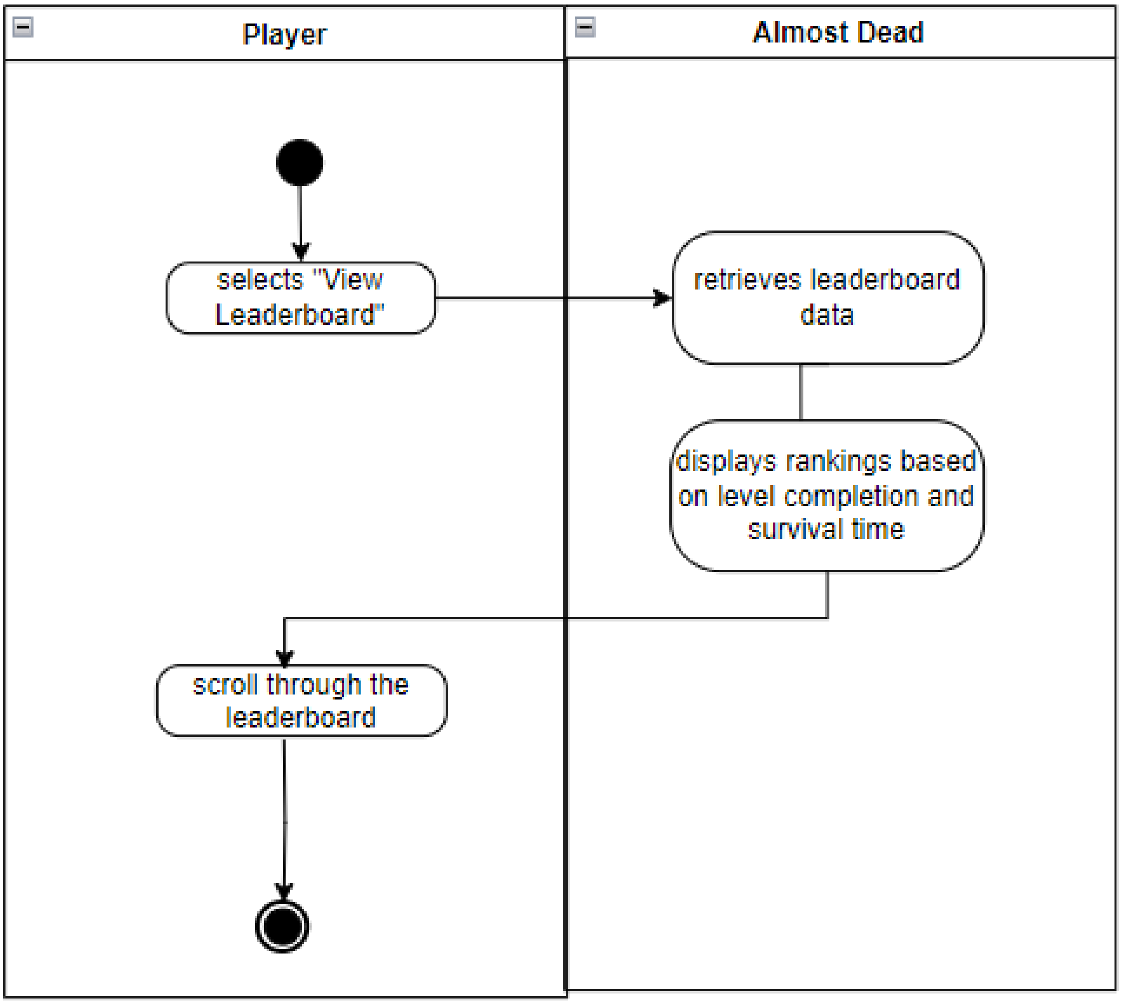

Figure 9.3: AD View Leader Board

9.4 Watch Intro Video

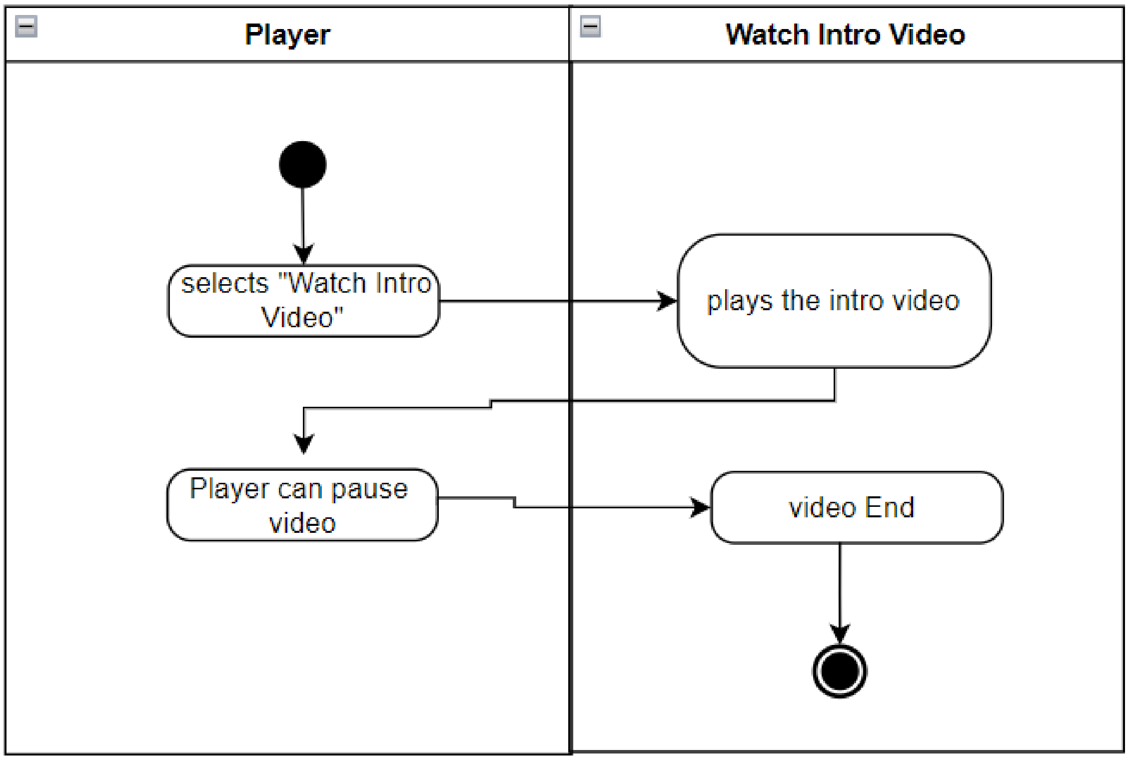

Figure 9.4: Ad Watch Intro Video

9.5 Resume/Pause Game

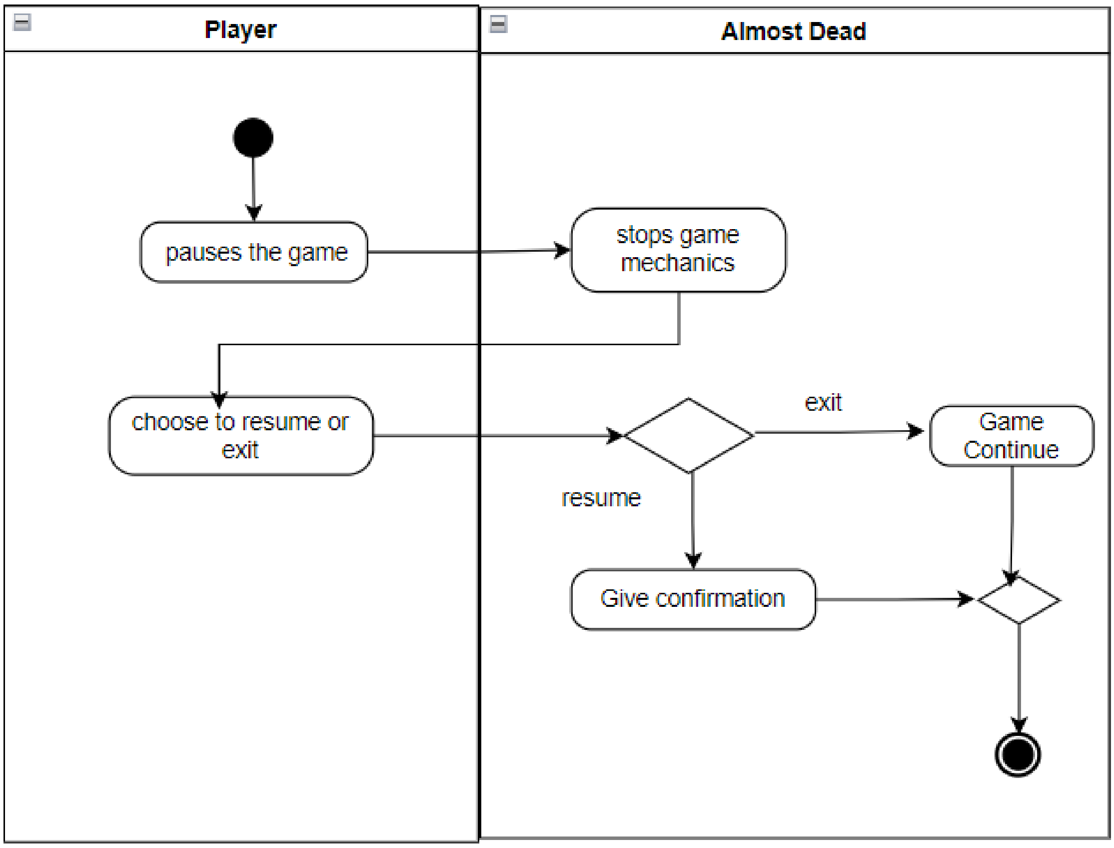

Figure 9.5: AD Resume/Pause Game

9.6 View User Manual

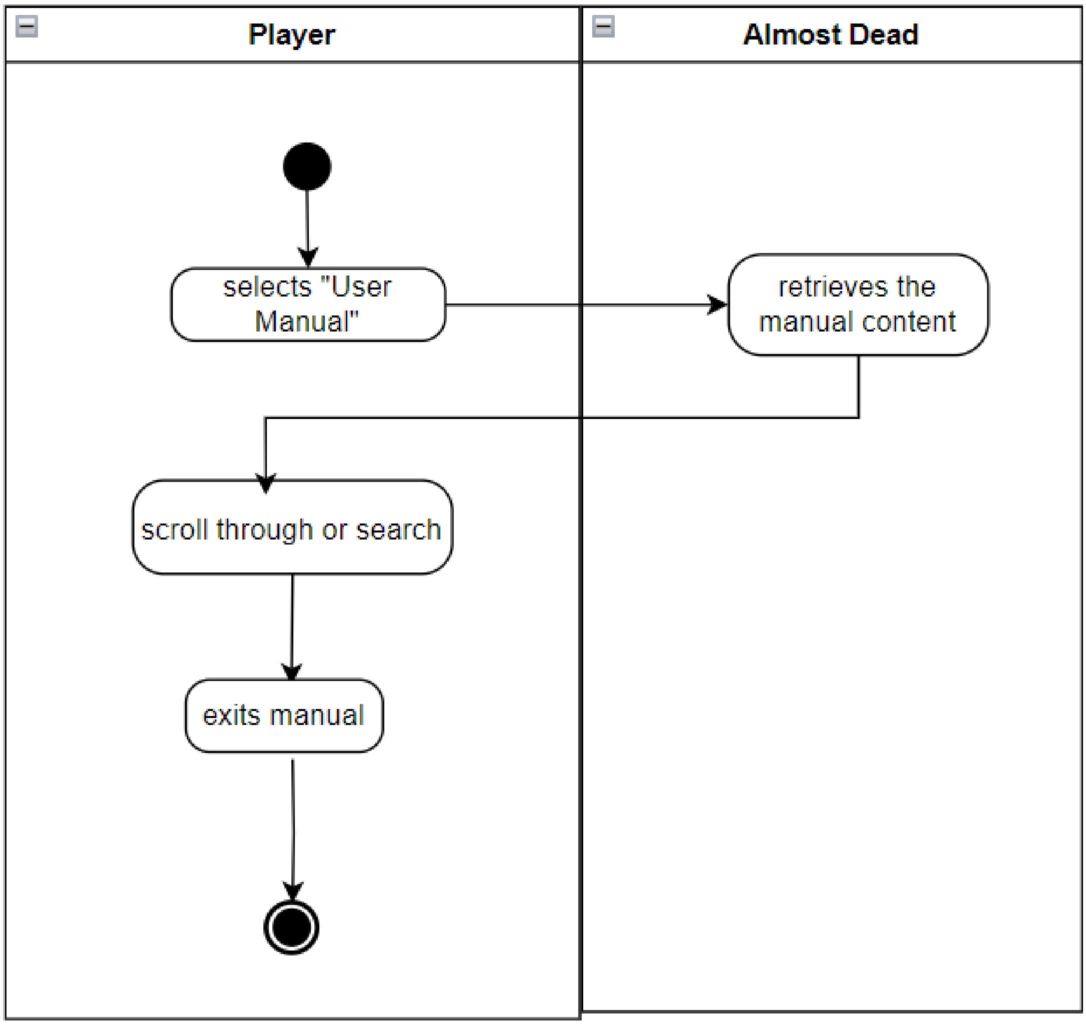

Figure 9.6: AD View User Manual

9.7 Play Game

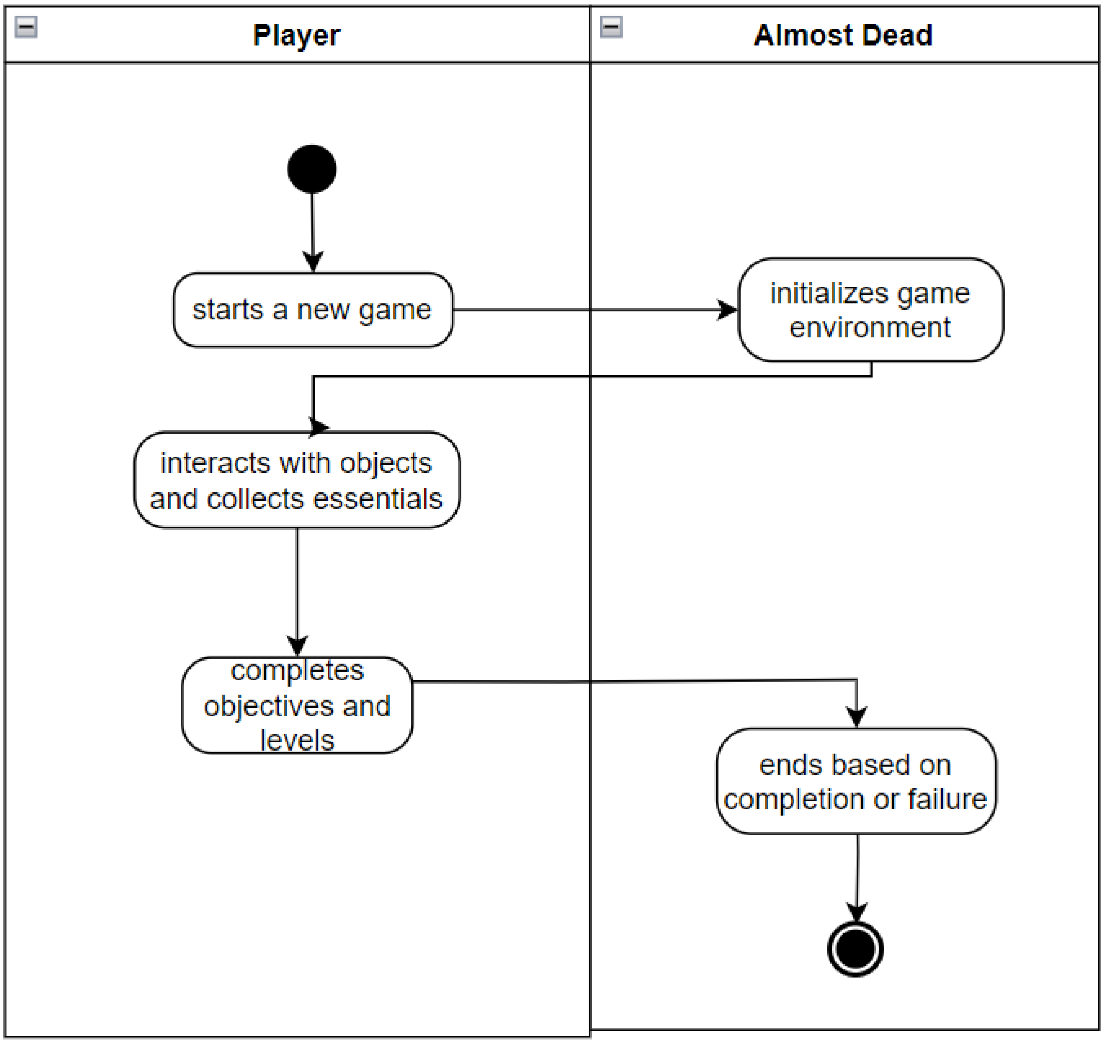

Figure 9.7: AD Play Game

9.8 Mini-Map Navigation

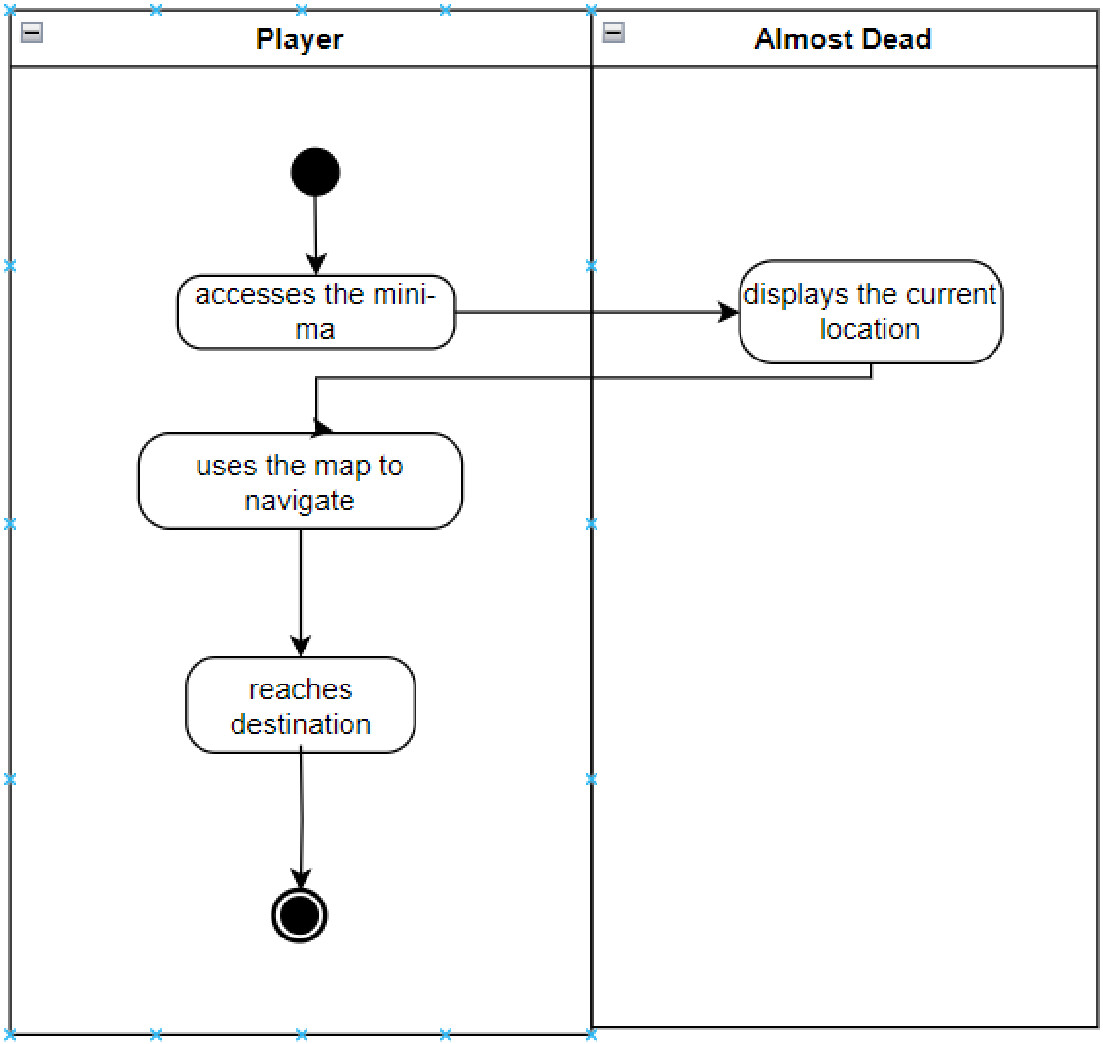

Figure 9.8: AD Mini-Map Navigation

9.9 Environment Navigation

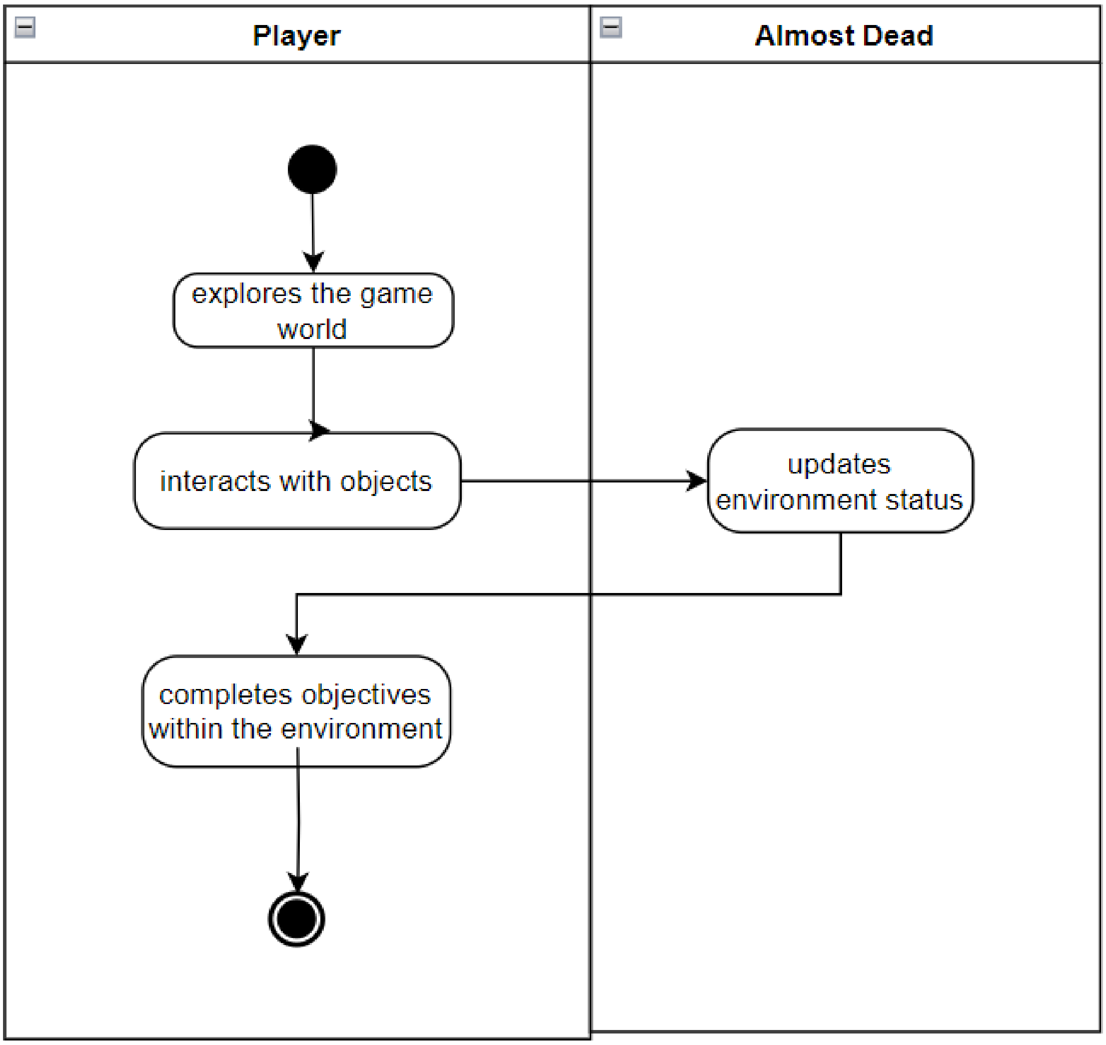

Figure 9.9: AD Environment Navigation

9.10 Collect Essential

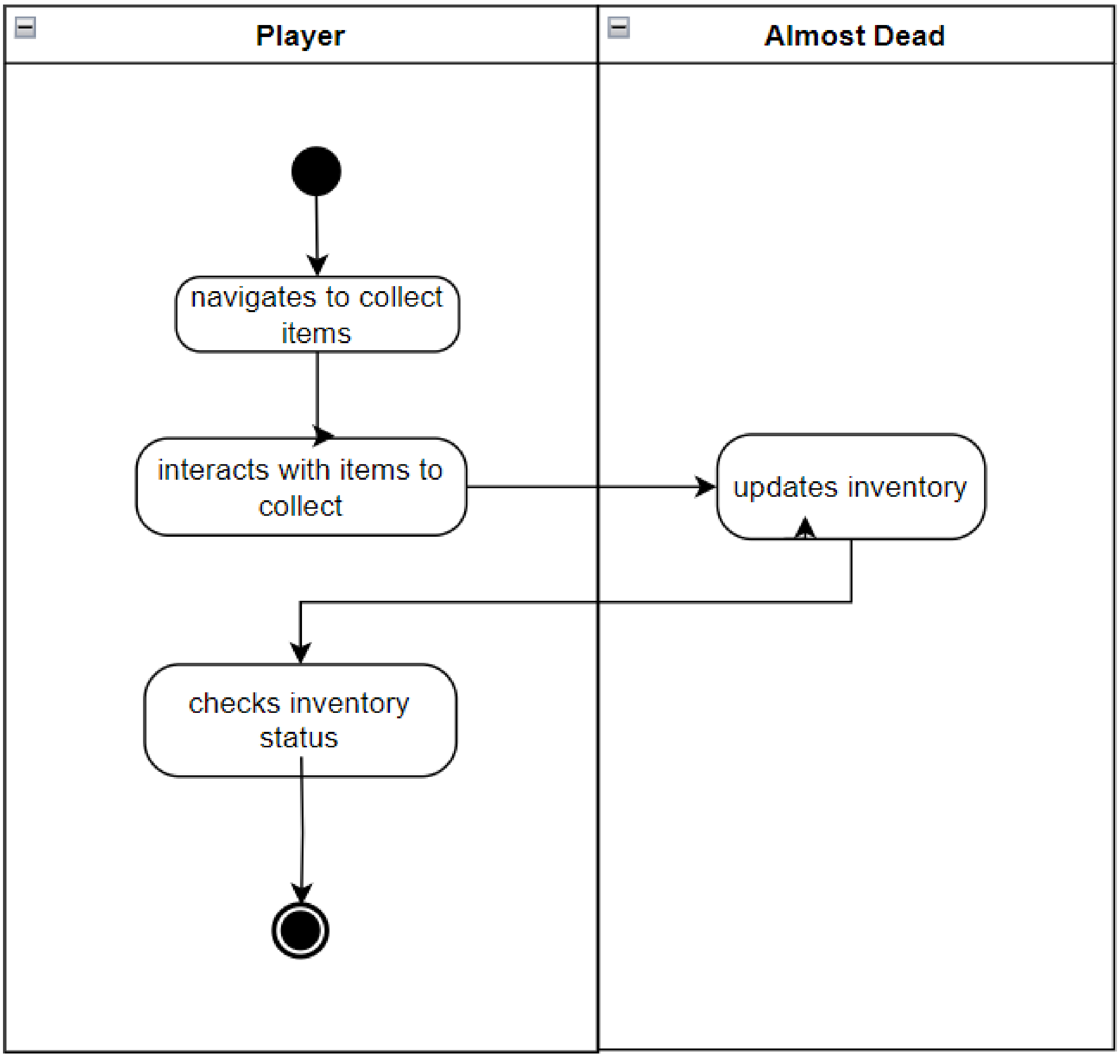

Figure 9.10: AD Collect Essential

9.11 Escape And Survival Mechanics

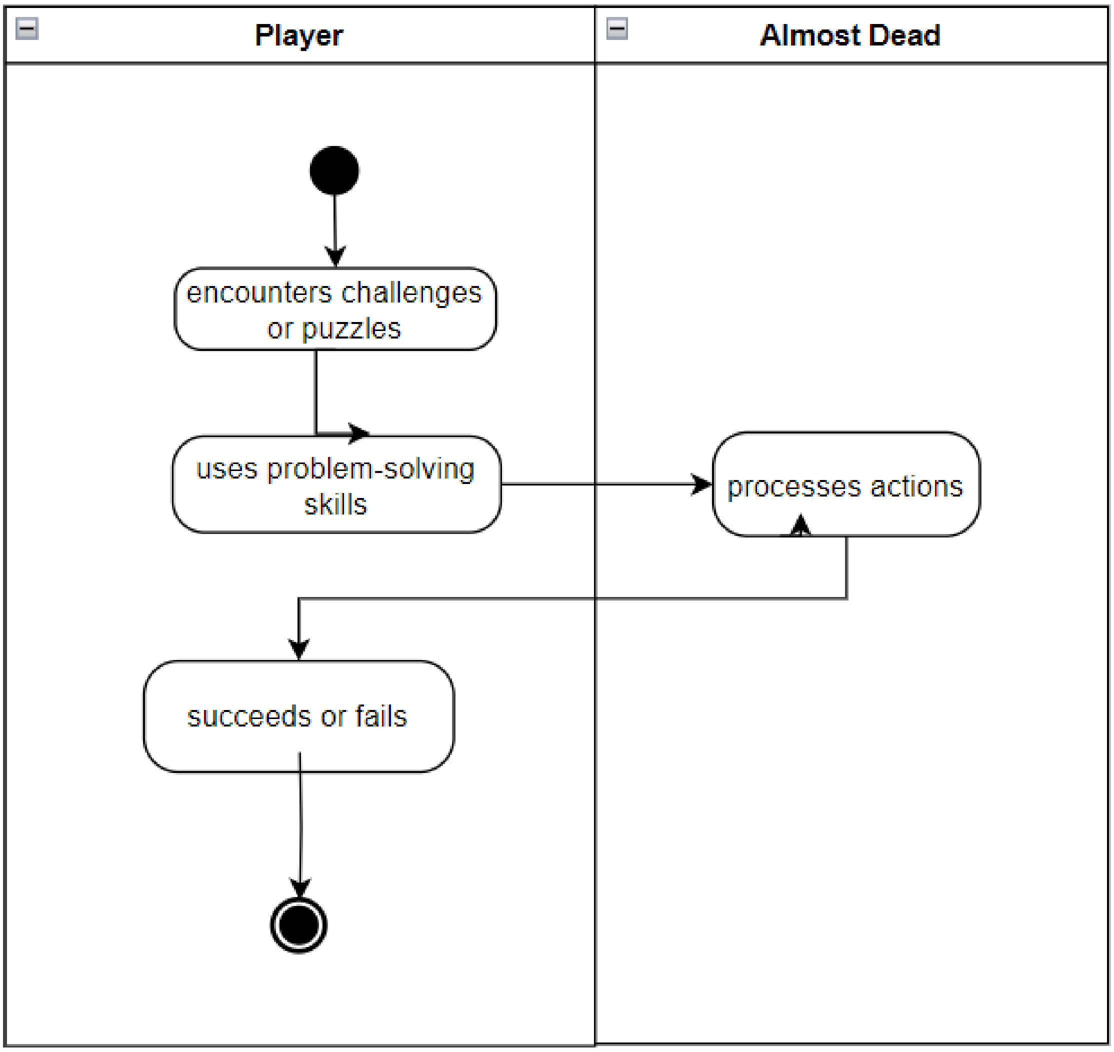

Figure 9.11: AD Escape And Survival Mechanics

9.12 Complete Level

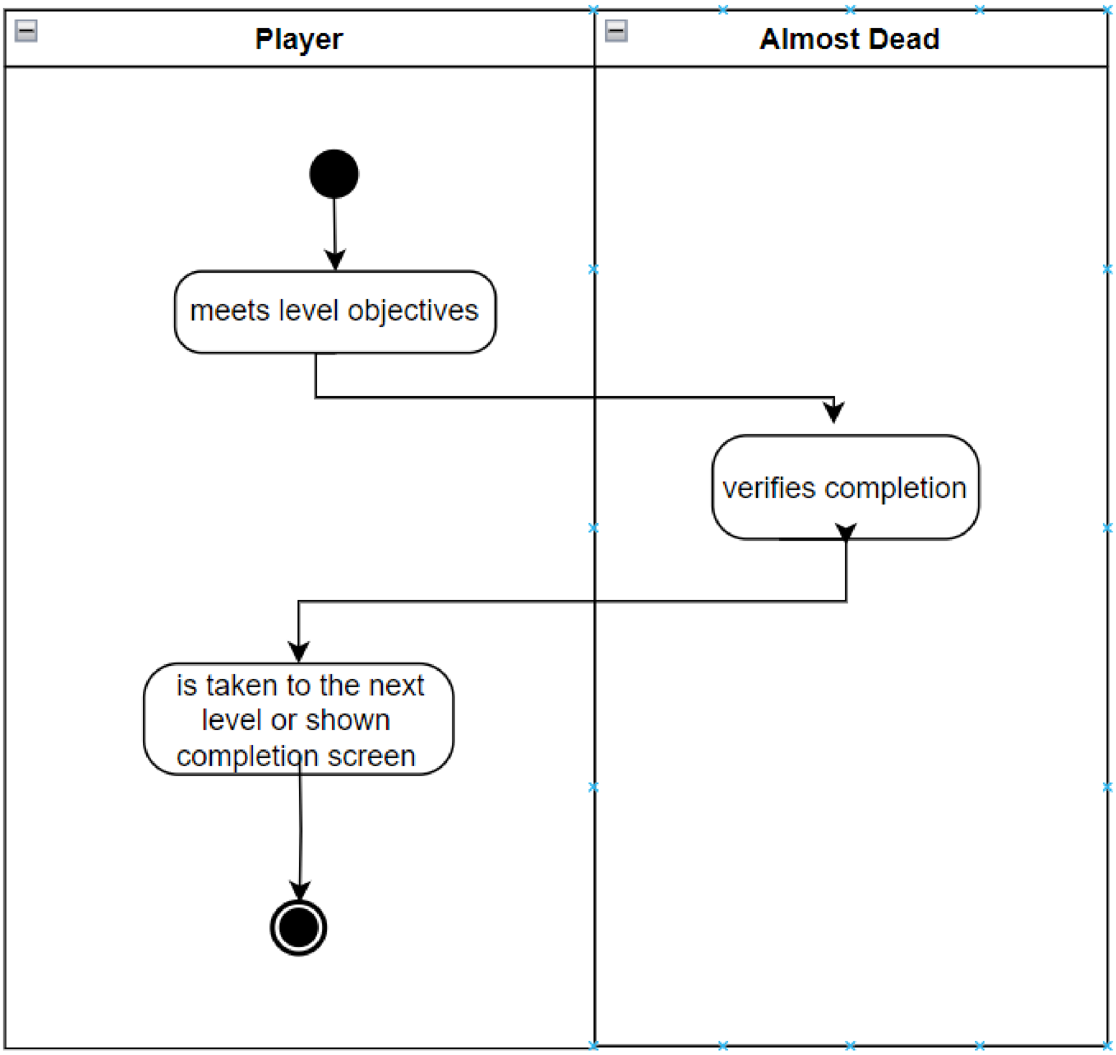

Figure 9.12: Complete Level

9.13 Gameplay Mechanics

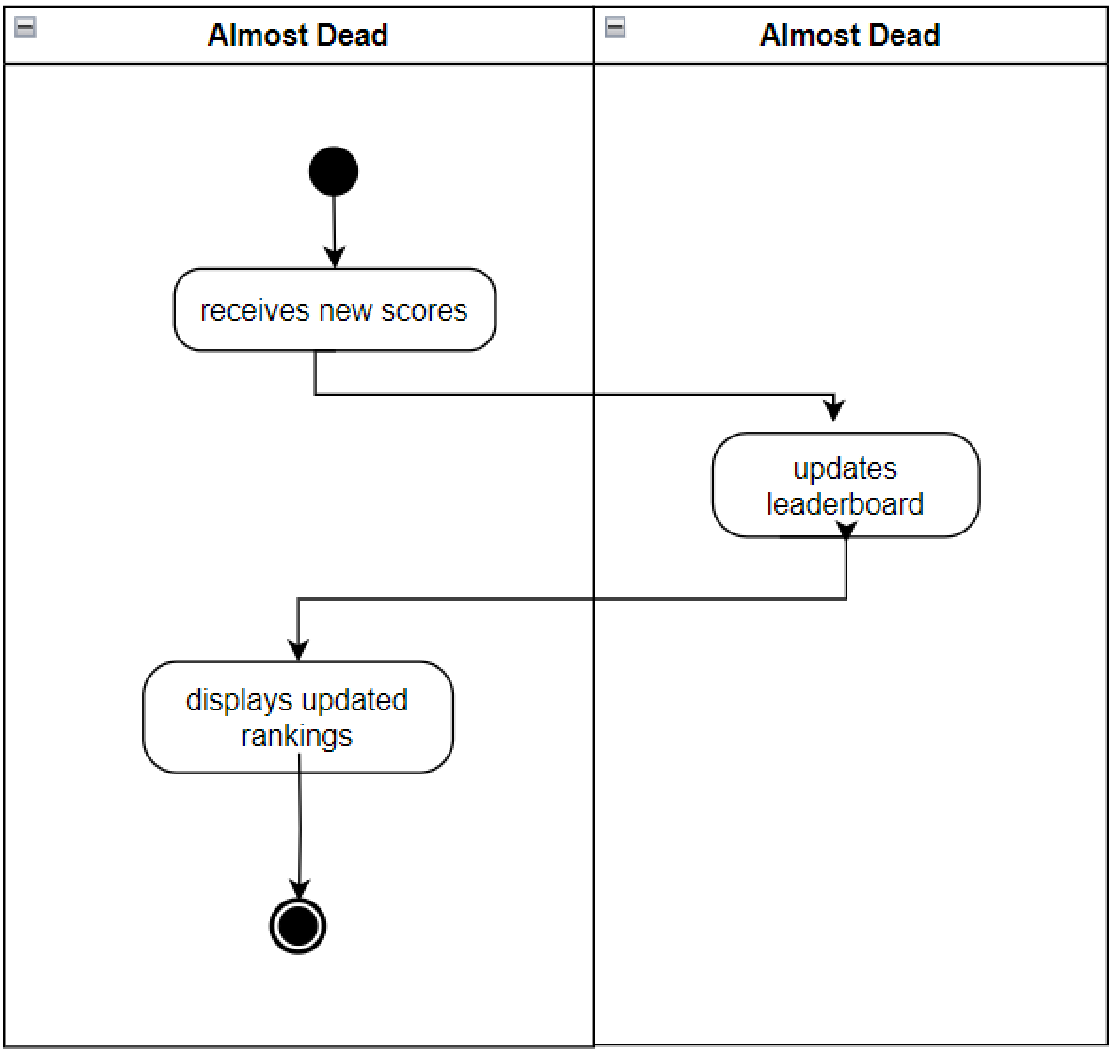

Figure 9.13: AD Gameplay Mechanics

9.14 Manager Leader Board

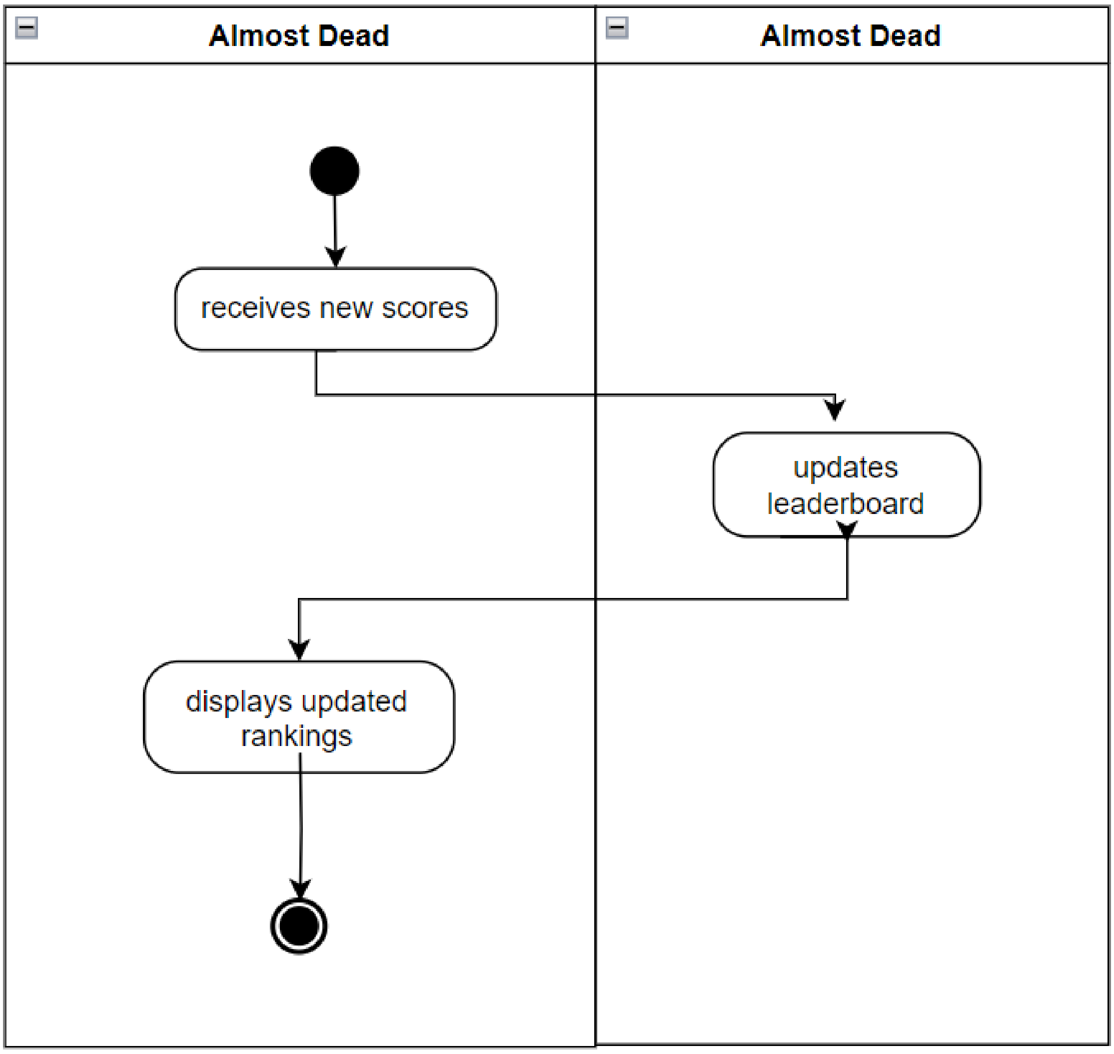

Figure 9.14: Ad Manager Leader Board

Chapter 10

Operations

10.1 Player Registration

[h!]

Operation:	enterRegistrationDetails(username, password, email)
Cross-references	Player Registration Use Case, User Authentication System, Firebase Authentication
Pre-condition	The game is installed on the player's device, and they are online. Firebase Authentication is integrated into the game system.
Post-condition	The account of the player is successfully created and safely kept. The email address the player registered with receives a confirmation email.

Table 10.1: Player Registration

10.2 User Login Management

Operation:	manageUserLogin(username, password)
Cross-references	User Login Management Use Case
Pre-condition	The player's account has already been made. The Firebase Authentication system is operational.
Post-condition	The player has logged in and been verified. The game progress of the player is loaded.

Table 10.2: User Login Management

10.3 Watch Intro Video

Operation:	playIntroVideo()
Cross-references	Watch Intro Video Use Case
Pre-condition	The video is available, and the game has begun. The video has not been skipped by the user.
Post-condition	The full introduction video is seen by the player. After the video, the game switches to the main menu.

Table 10.3: Watch Intro Video

10.4 View Leaderboard

Operation:	requestLeaderboard()
Cross-references	View Leaderboard Use Case
Pre-condition	The player has been authenticated. The leaderboard data is current and readily available.
Post-condition	A leaderboard with precise rankings is displayed to the player.

Table 10.4: View Leaderboard

10.5 Resume/Pause Game

Operation:	pauseGame()
Cross-references	Resume/Pause Game Use Case
Pre-condition	The game is currently being played.
Post-condition	The game is paused, and the player picks up where they left off.

Table 10.5: Resume/Pause Game

10.6 View User Manual

Operation:	viewManual()
Cross-references	View User Manual Use Case
Pre-condition	The in-game manual is available to the player who is currently logged in.
Post-condition	The guide can be successfully viewed by the player.

Table 10.6: View User Manual

10.7 Play Game

Operation:	startGame()
Cross-references	Play Game Use Case
Pre-condition	After logging in, the player chose to "Play."
Post-condition	The game world is successfully navigated by the player as they advance through the levels.

Table 10.7: Play Game

10.8 Mini-Map Navigation

Operation:	displayMiniMap()
Cross-references	Mini-Map Navigation Use Case
Pre-condition	The player must keep track of targets or locations. In-game, the player can access the mini-map.
Post-condition	Using the mini-map, the player successfully navigates the surroundings. There is accurate display of the locations and goals.

Table 10.8: Mini-Map Navigation

10.9 Environment Navigation

Operation:	PlayerMovement(dir, sp)
Cross-references	Environment Navigation Use Case
Pre-condition	There are goals or areas for the player to investigate in the game
Post-condition	The player navigates the environment with ease and without any problems. There is accessibility to all important areas and goals.

Table 10.9: Environment Navigation

10.10 Collect Essentials

Operation:	collectItem(playerID, itemID, distance)
Cross-references	Collect Essentials Use Case
Pre-condition	There are collectible items in the level the player is in. Everything is set up and prepared for pickup.
Post-condition	The player succeeds in gathering every necessary item. In the player's inventory, the items they have collected are added.

Table 10.10: Collect Essentials

10.11 Escape and Survival Mechanics

Operation:	activateSurvivalMechanism(playerID, mechanismType)
Cross-references	Escape and Survival Mechanics Use Case
Pre-condition	The player is actively avoiding dangers and navigating the surroundings. The system has offered chances for escape or obstacles for survival.
Post-condition	By successfully avoiding danger, the player completes the level. The player gets away to the next area of safety or level.

Table 10.11: Escape and Survival Mechanics

10.12 Complete Level

Operation:	AllObjectiveDone()
Cross-references	Complete Level Use Case
Pre-condition	The player has finished all of the level's prerequisite tasks.
Post-condition	After completing the current level successfully, the player advances to the next. The game keeps track of your progress and lets you access the next level.

Table 10.12: Complete Level

10.13 Manage Leaderboard

Operation:	AnalyzeUserPerformance()
Cross-references	Manage Leaderboard Usercase
Pre-condition	For leaderboard ranking, player performance data is available. A level has been completed by the player.
Post-condition	The player's ranking and score are updated on the leaderboard. The leaderboard displays the latest results.

Table 10.13: Manage Leaderboard

10.14 Gameplay Mechanics

Operation:	GameMechanics(interact,explore,survive)
Cross-references	Gameplay Mechanics Use Case
Pre-condition	Gameplay mechanics are enabled, and the player is in the game
Post-condition	The game accurately reflects the player's actions, and the gameplay mechanics are executed with ease.

Table 10.14: Gameplay Mechanics

Chapter 11

Class Diagram

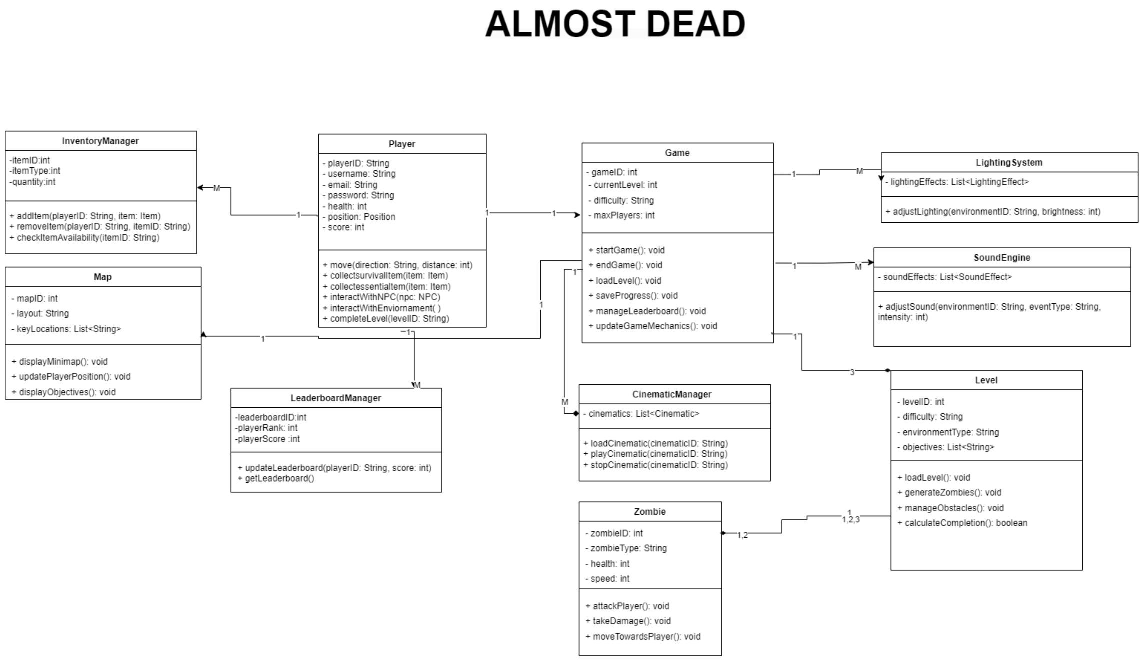

Figure 11.1: Class Diagram

Chapter 12

Sequence Diagram

12.1 Watch Intro Video

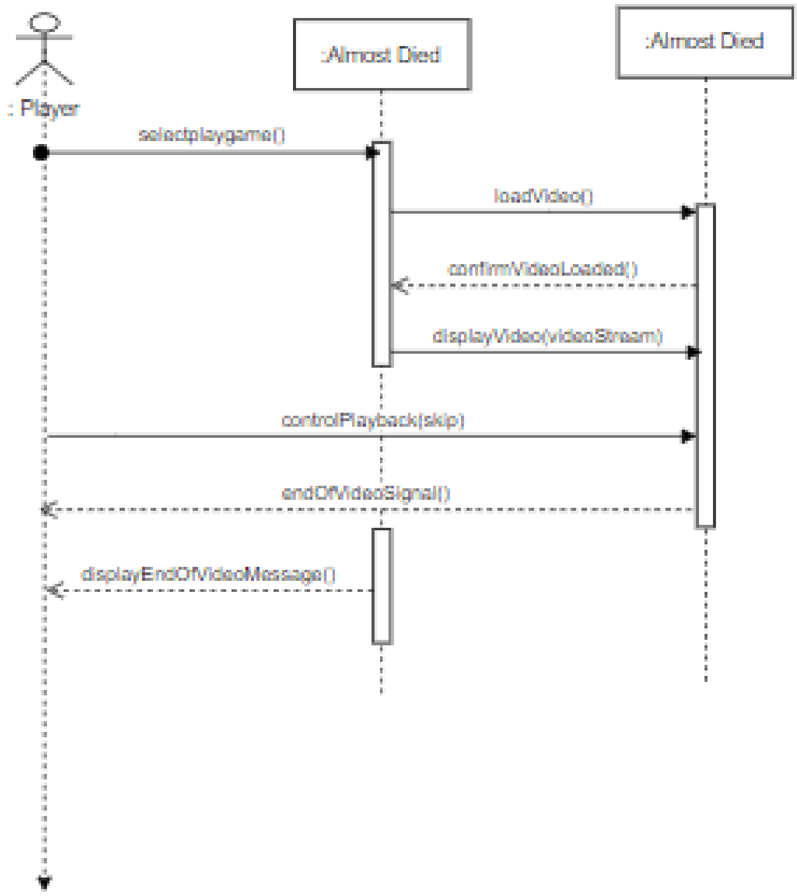

Figure 12.1: SD Watch Intro Video

12.2 Player Registration

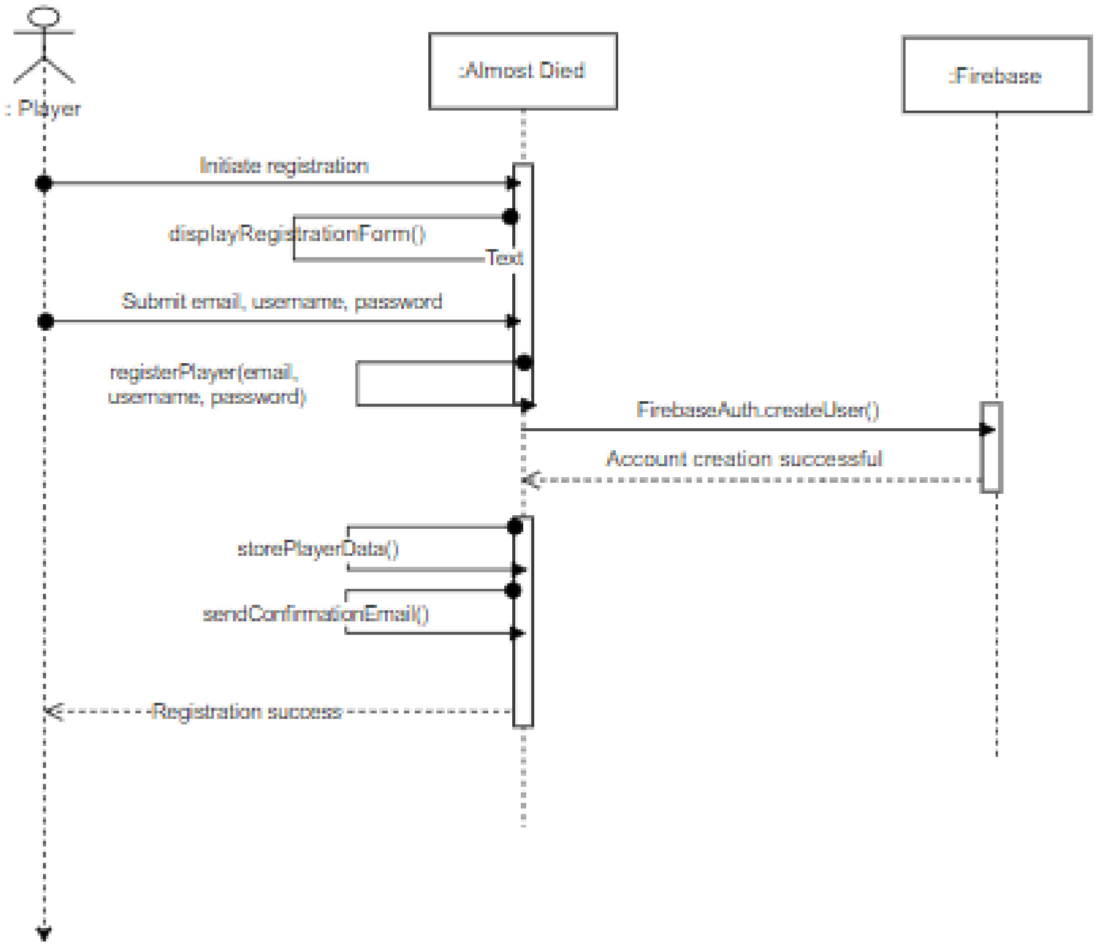

Figure 12.2: SD Player registration

12.3 Game Play Mechanics

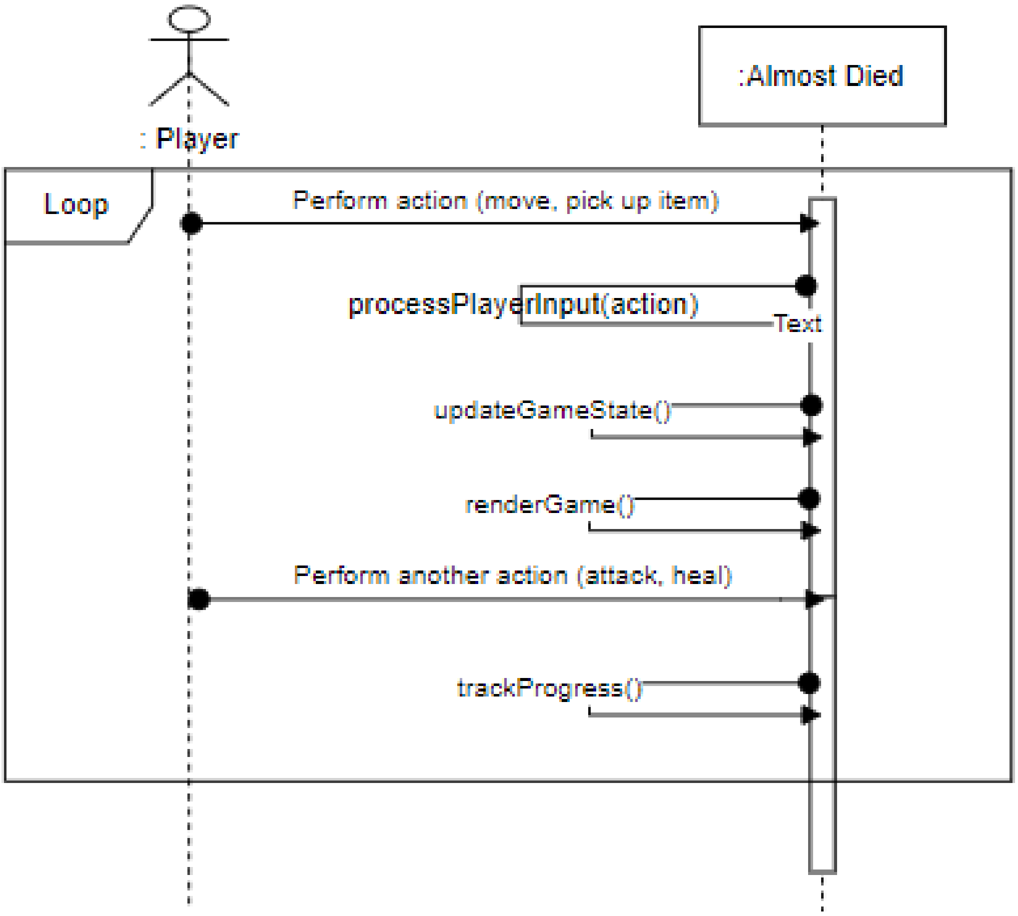

Figure 12.3: SD Game Play Mechanics

12.4 Environment Navigation

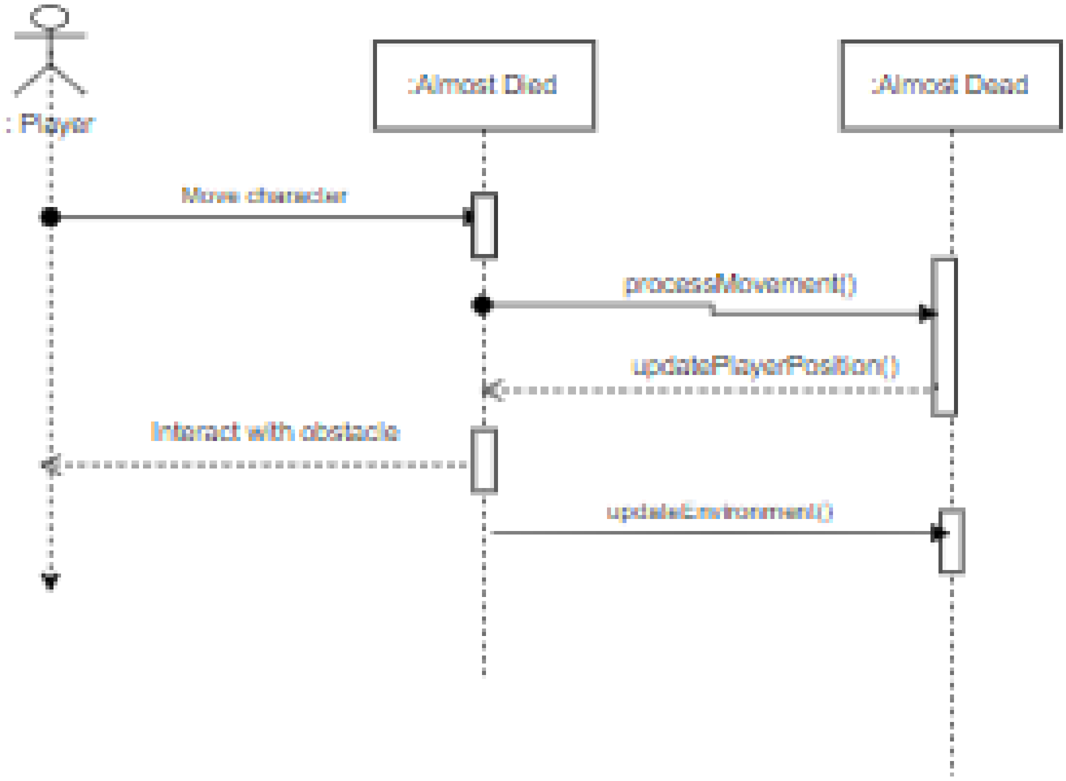

Figure 12.4: SD Environment Navigation

12.5 MiniMap Navigation

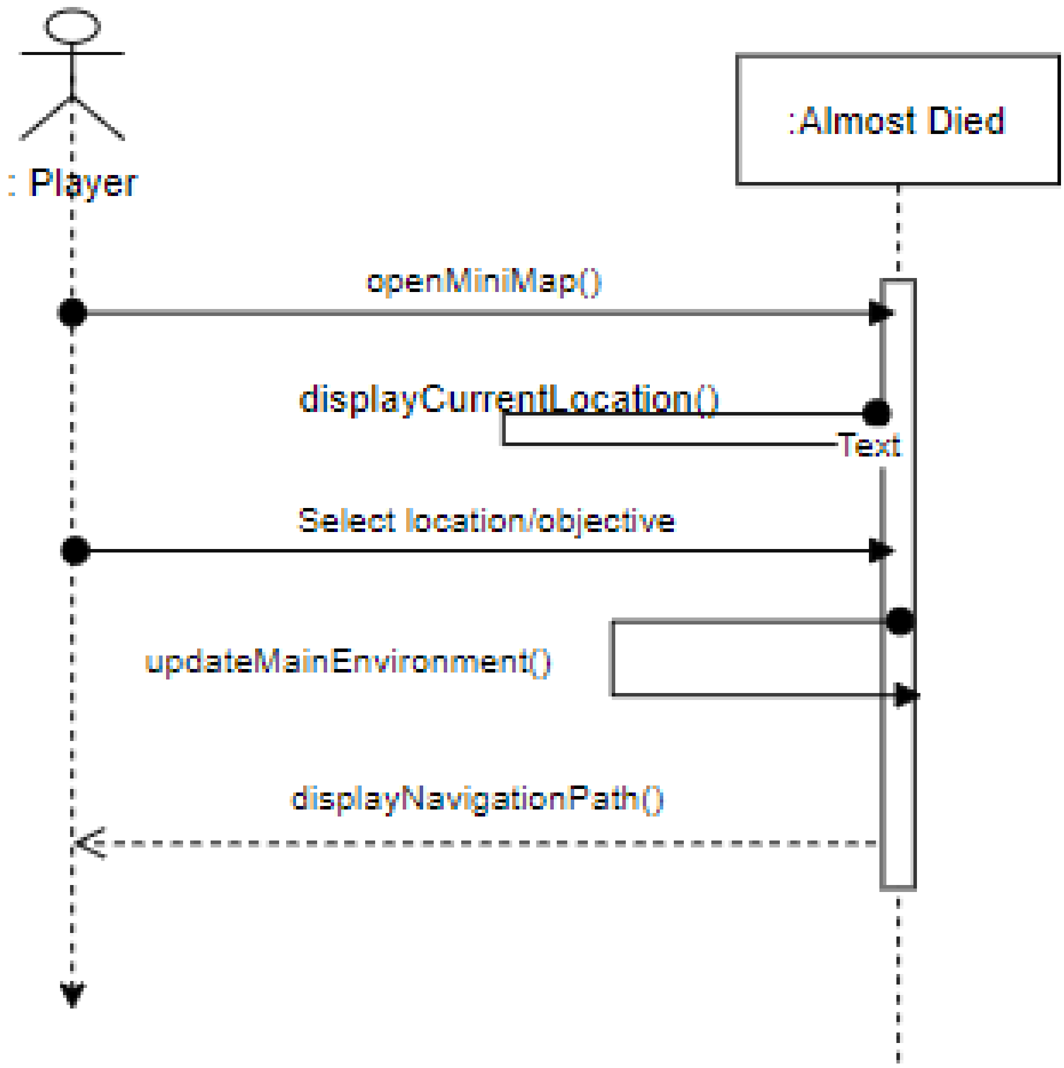

Figure 12.5: SD Minimap Navigation

12.6 View User Manual

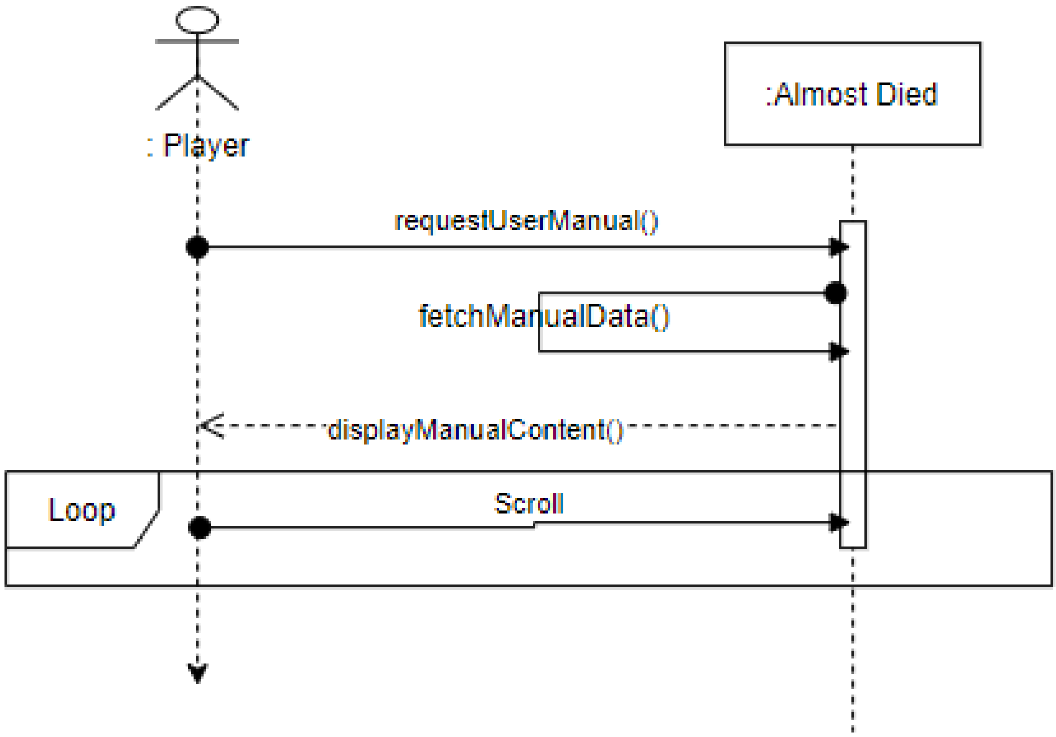

Figure 12.6: SD View User Manual

12.7 Resume/Pause Game

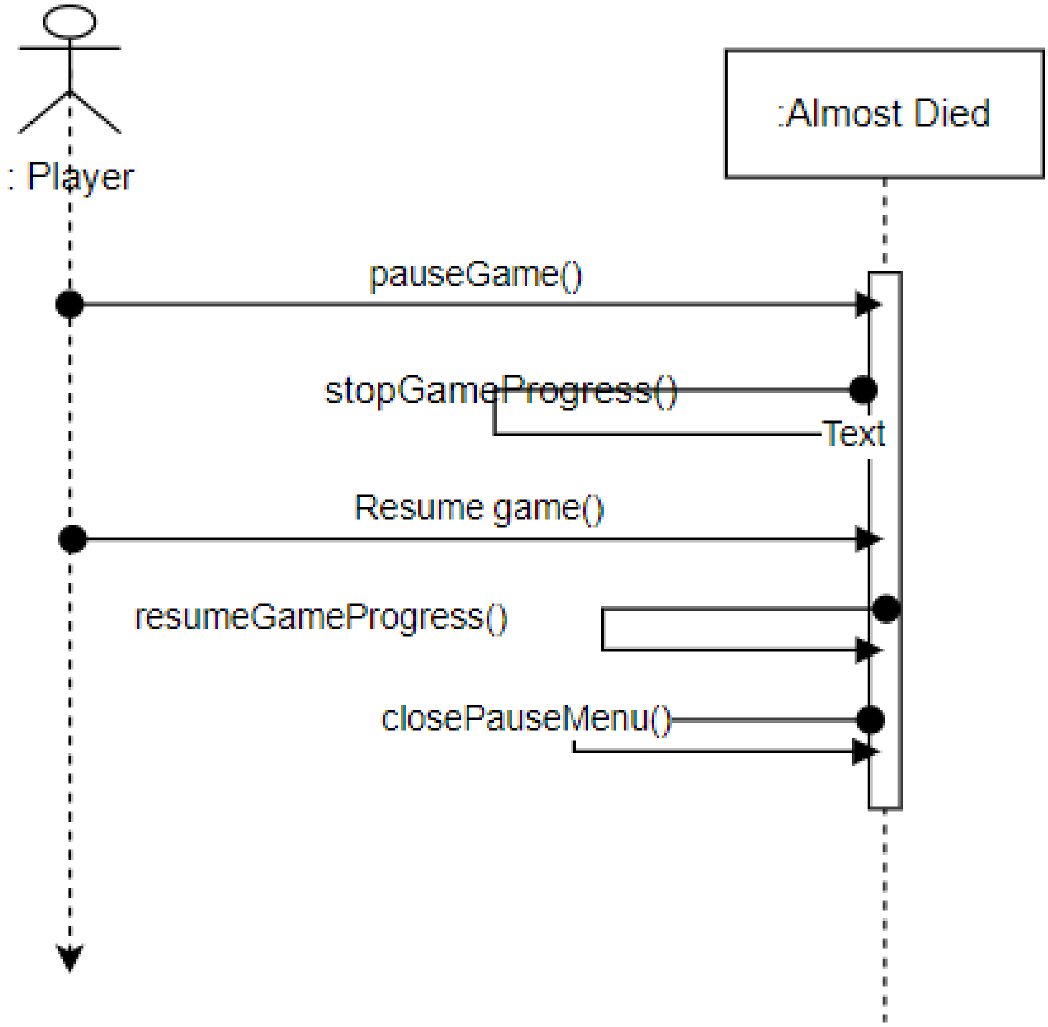

Figure 12.7: SD Resume/Play Game

Chapter 13

ER Diagram

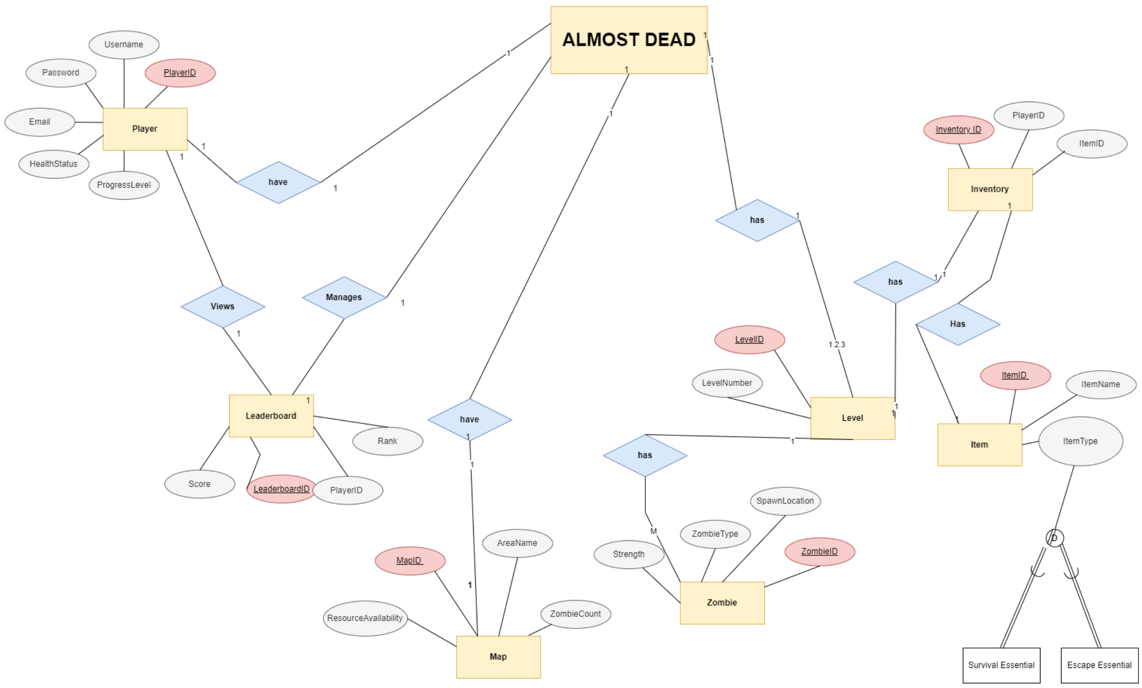

Figure 13.1: ERD

Chapter 14

Architecture Diagram

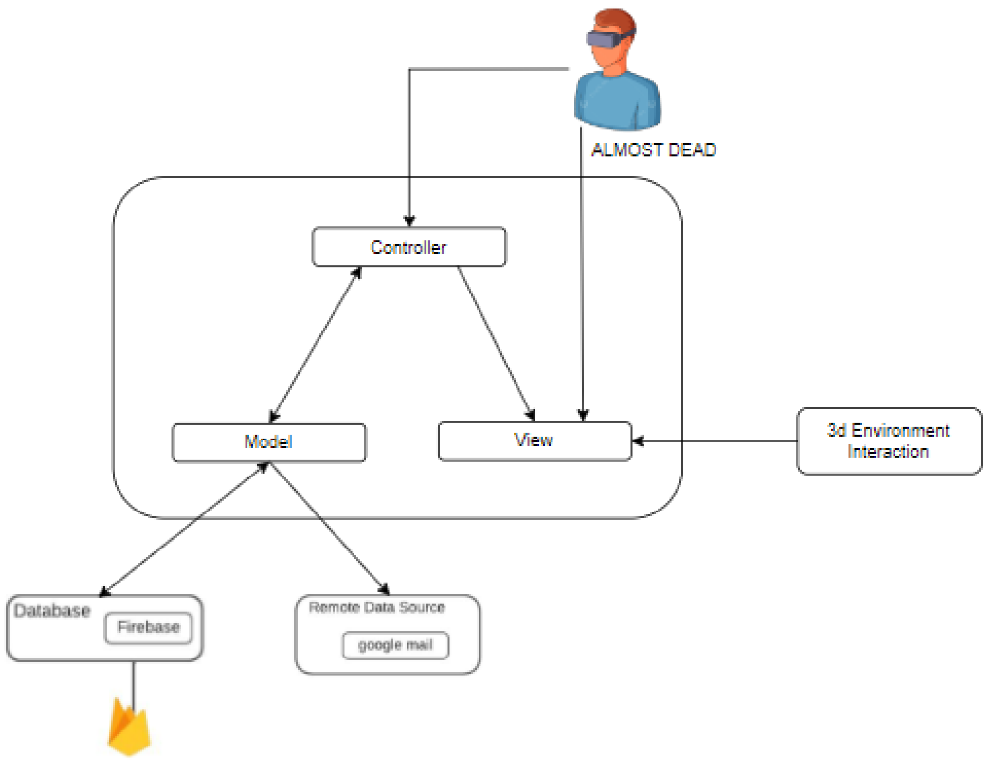

Figure 14.1: Architecture Diagram

Chapter 15

Conclusion

The "Almost Dead" VR zombie survival game's Software Requirements Specification (SRS) document provides a thorough framework for the design, development, and implementation of the game. This document includes several UML diagrams, such as use case, sequence, entity-relationship, and activity diagrams, along with thorough descriptions of all functional and non-functional requirements. These graphics depict the relationships between important entities, the data flow in the game, and the interactions between players and system elements.

To ensure that all stakeholders have a common understanding of the goals, gameplay mechanics, and anticipated user experiences of the game, this report is an essential resource. "Almost Dead" seeks to provide a distinct and captivating experience by stressing user engagement through immersive environments, non-violent gameplay, and strategic problem-solving. For developers and designers to build a compelling and dynamic virtual reality game that players will enjoy, the requirements and diagrams provided offer a strong starting point.

Bibliography